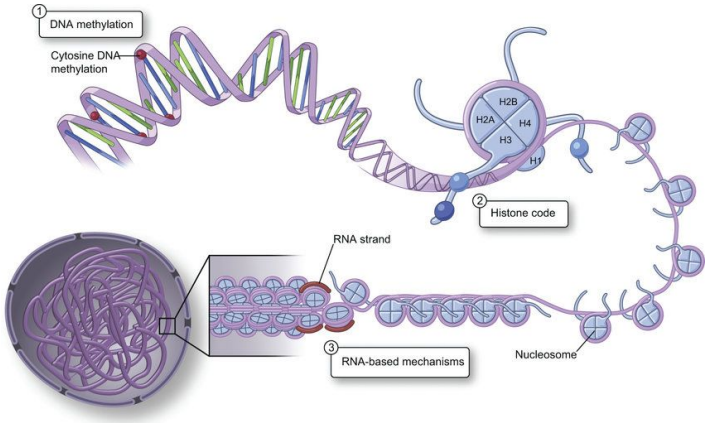
BME 105: Genetics in the Genomics Era

Lecture 14: Gene Regulation and Epigenetics



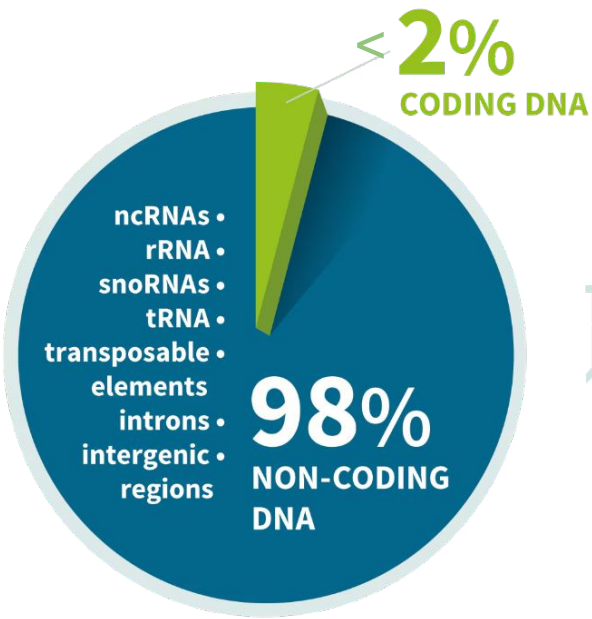
Instructor: Dr. Shelbi Russell,
Assistant Professor

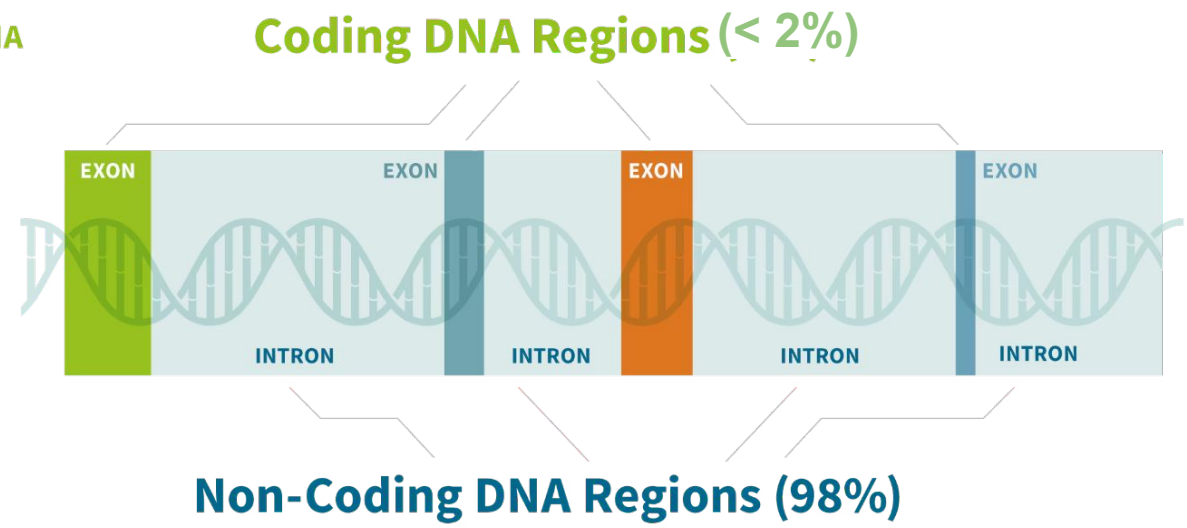
Course website on canvas.ucsc.edu

Lecture 14 Learning Objectives

- Understand the importance of the non-coding regions of the genome: where do most of the trait and disease-linked SNPs reside?
- Be able to define epigenetics: heritable alteration in gene expression <u>not due to mutation</u> in DNA sequence
- Describe epigenetic regulation at the DNA level, i.e., DNA methylation, nucleosome packaging, and histone modification (methylation/acetylation)
- Describe epigenetic regulation at the transcriptional level
  - Distinguish cis- and trans-level transcriptional regulation, and what elements contribute to each category.
  - Describe the general ways transcriptional activators and repressors work (chromatin and transcriptional impacts)
    - Be able to describe the function of a transcription factor - does it always induce transcription?
- Begin to understand how the genome uses short and long non-coding RNAs for epigenetic regulation.
- Understand the basic sequencing approaches for evaluating transcriptional regulation at the chromatin level: e.g., ChIP-seq, Hi-C, ATAC-seq
- Appreciate that sequencing assays for associating genotypes with phenotypes computationally/statistically need to be validated in vivo/vitro.

Most of the human genome is non-coding

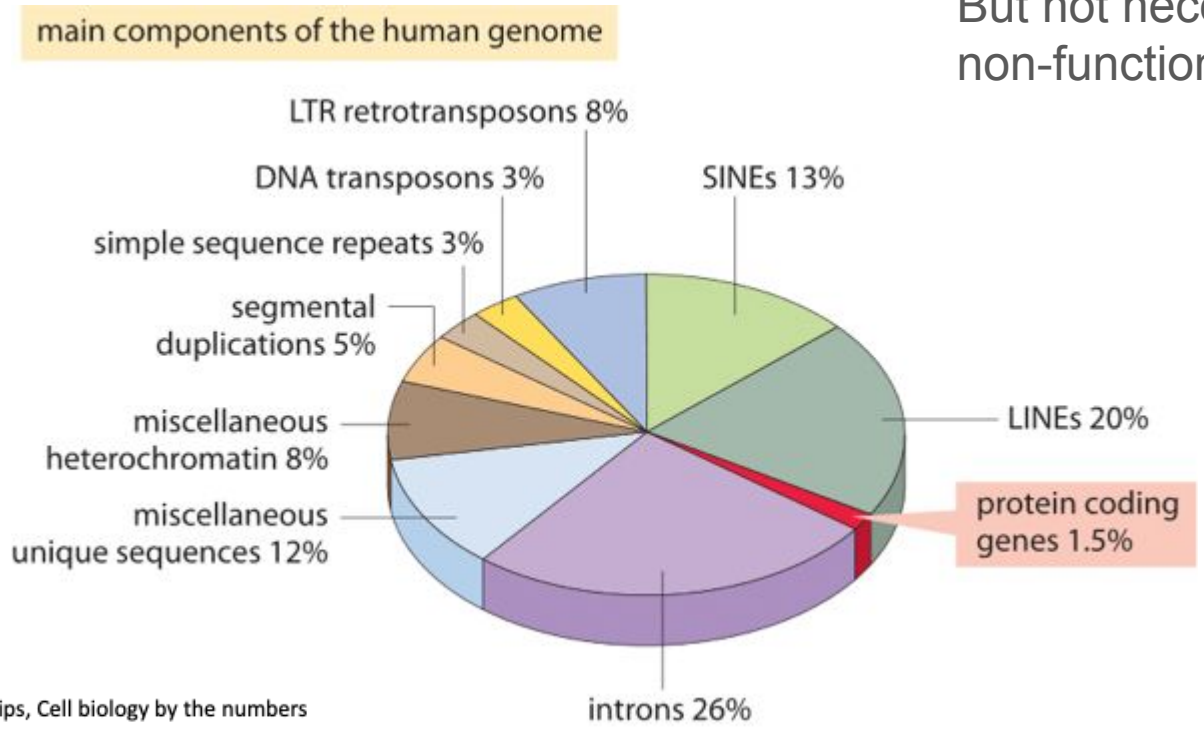




https://www.ancestry.com/c/dna-learning-hub/non-coding-dna

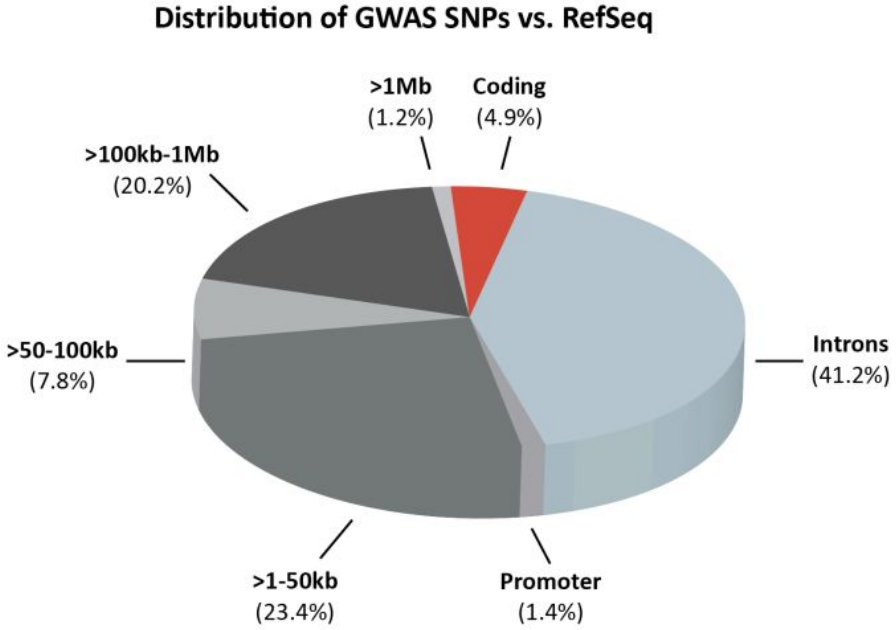
Most of the human genome is non-coding

But not necessarily non-functional…



Gregory 2005; Milo and Phillips, Cell biology by the numbers

Most human traits and diseases are associated with SNPs in non-coding regions

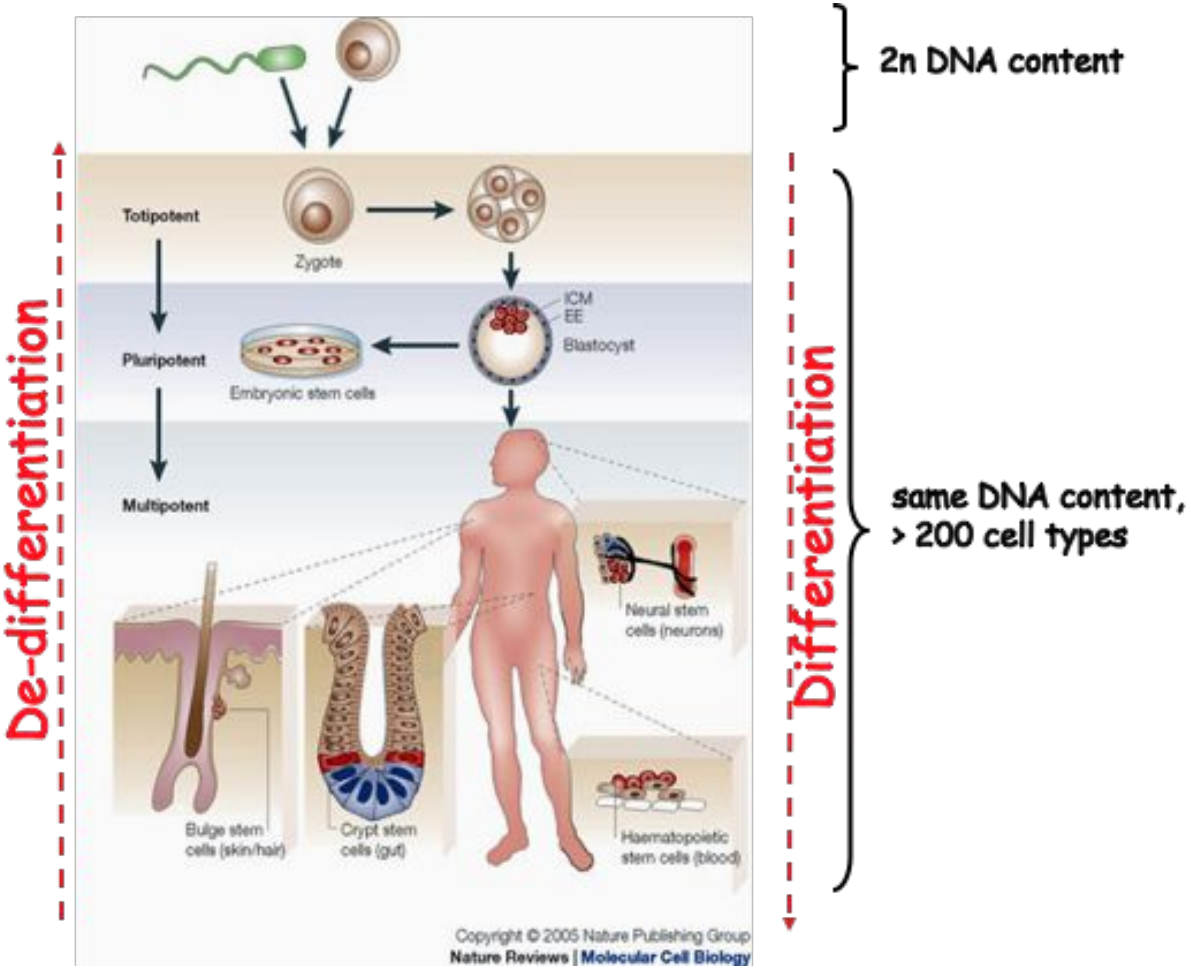


Maurano et al. Science 2012

What do non-coding regions do?

Non-coding regions of the genome help control what genes are turned on and off to give specialized cellular function spatially and temporally.

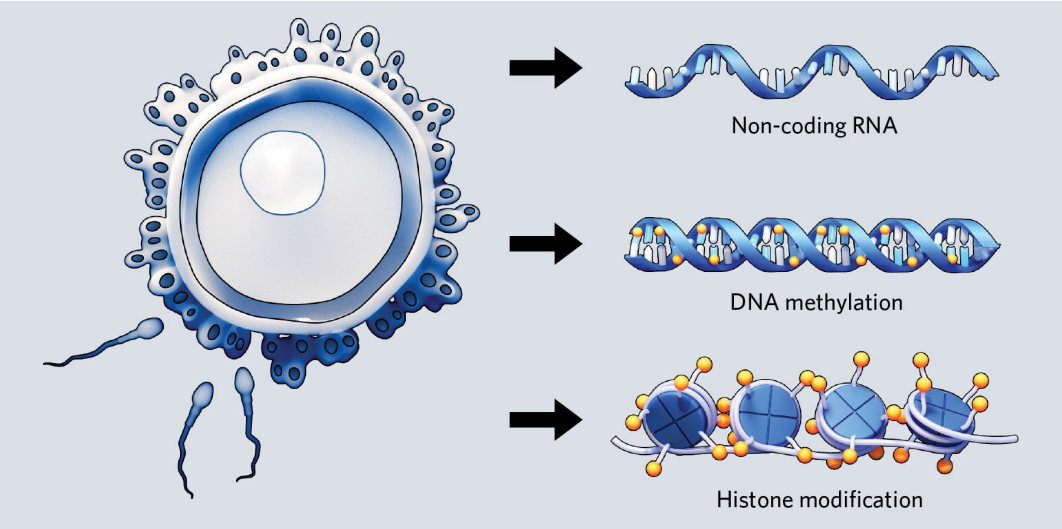
⇒ Non-coding regions control expression of the genome!



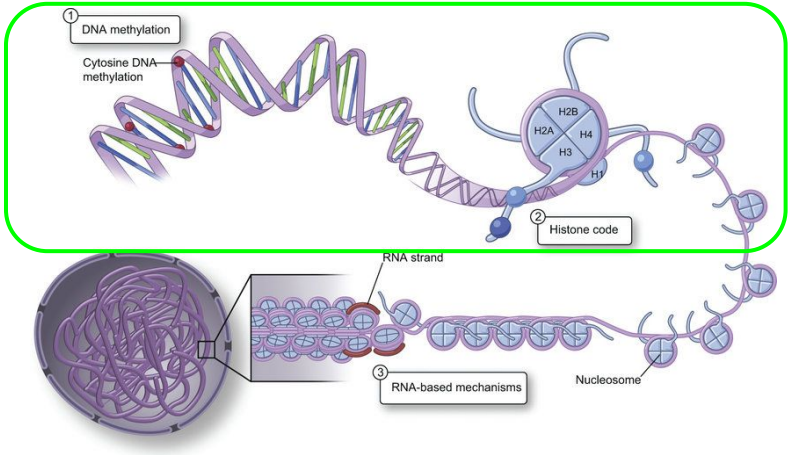
Epigenetics

- **Definition: heritable alteration in gene expression <u>not</u> due to mutation in DNA sequence**

  ⇒ A stable change in DNA packaging and accessibility
- Gene regulation through **stable** activation/repression of gene expression
- This is essential for normal cell differentiation and development of an organism after fertilization

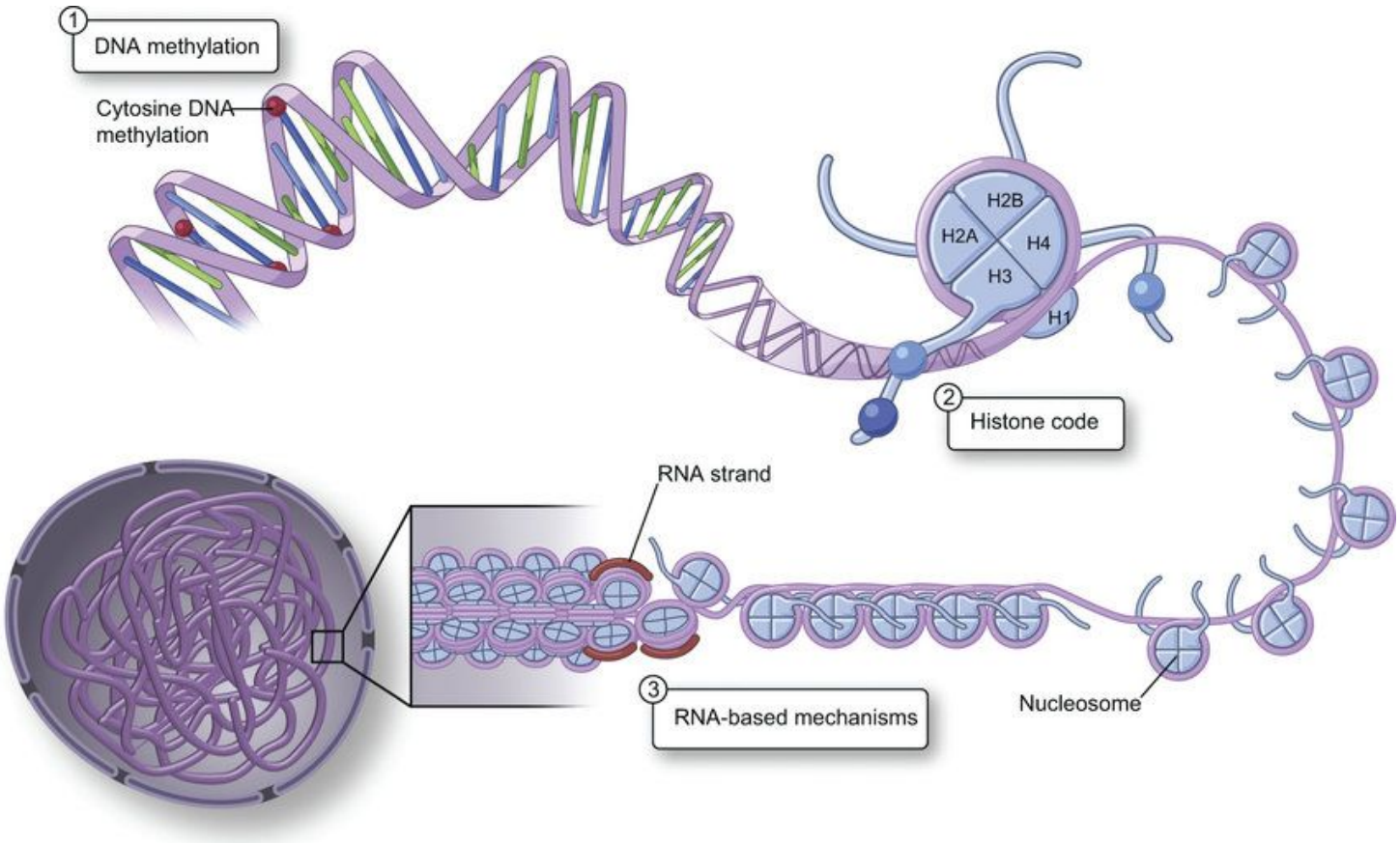


How are genes regulated?



How are genes regulated?

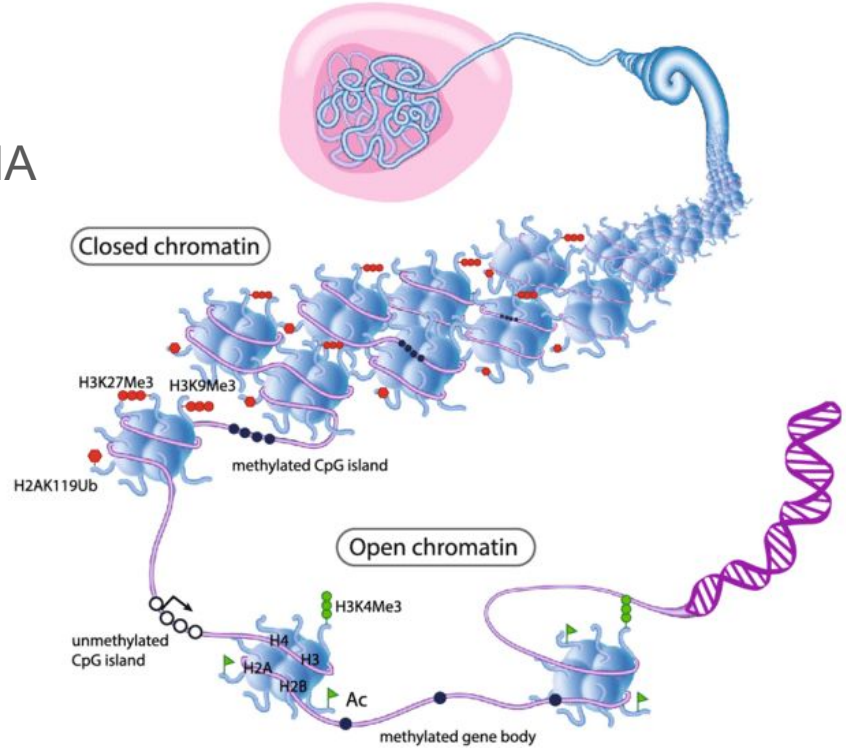
1. At the chromatin level - regulating access to the DNA

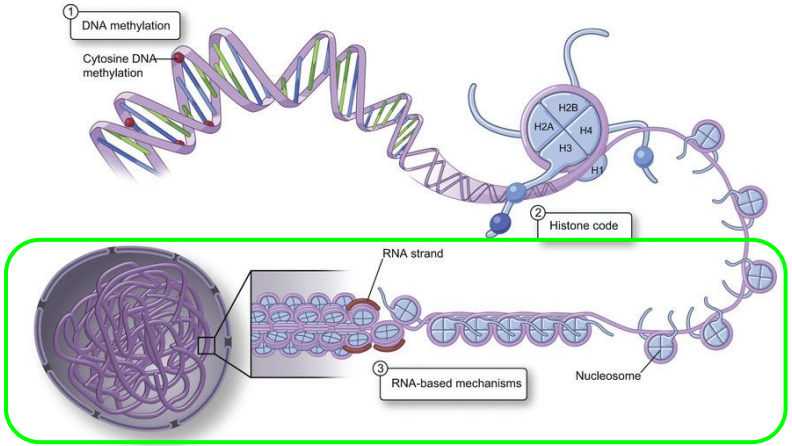


Yan et al. 2010. Journal of Applied Physiology

How do non-coding regions control coding regions?

1. At the chromatin level - regulating access to the DNA





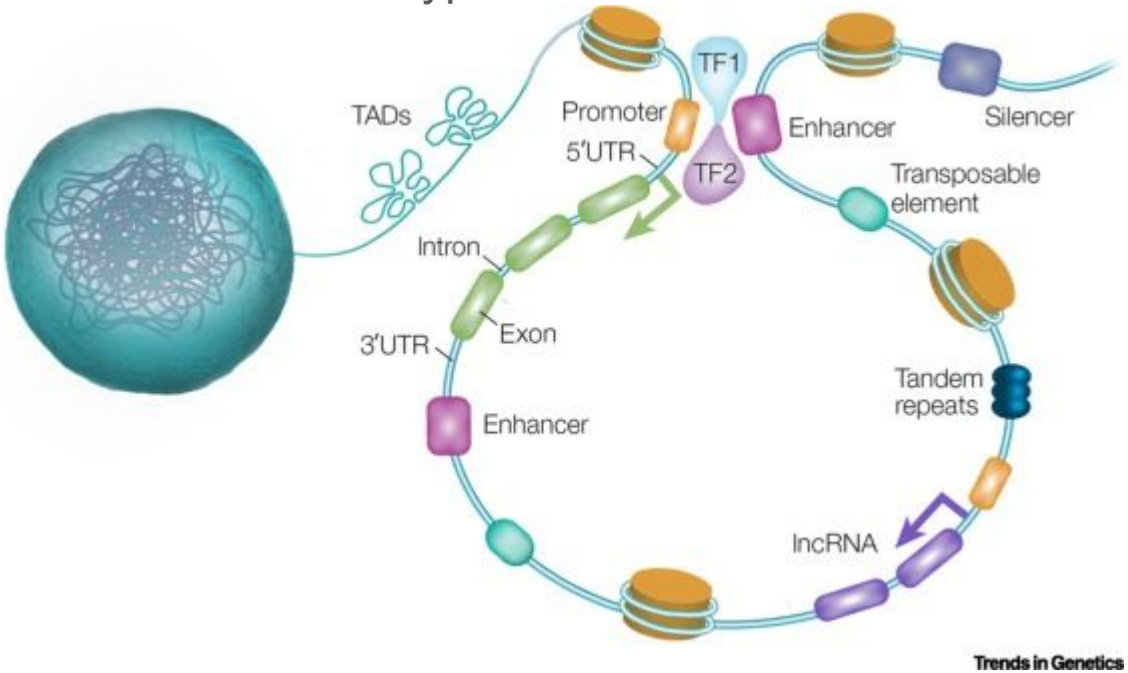
How are genes regulated epigenetically?

1. At the chromatin level - regulating access to the DNA
2. At the transcriptional level - modulating the amount and type of RNA

(both of these modifications can be transmitted during cell division)

How do non-coding regions control coding regions?

2. At the transcriptional level - modulating the amount and type of RNA



- Non-coding regions <u>contain</u>:
  - Promoters
  - Enhancers
  - Silencers
  - ncRNAs

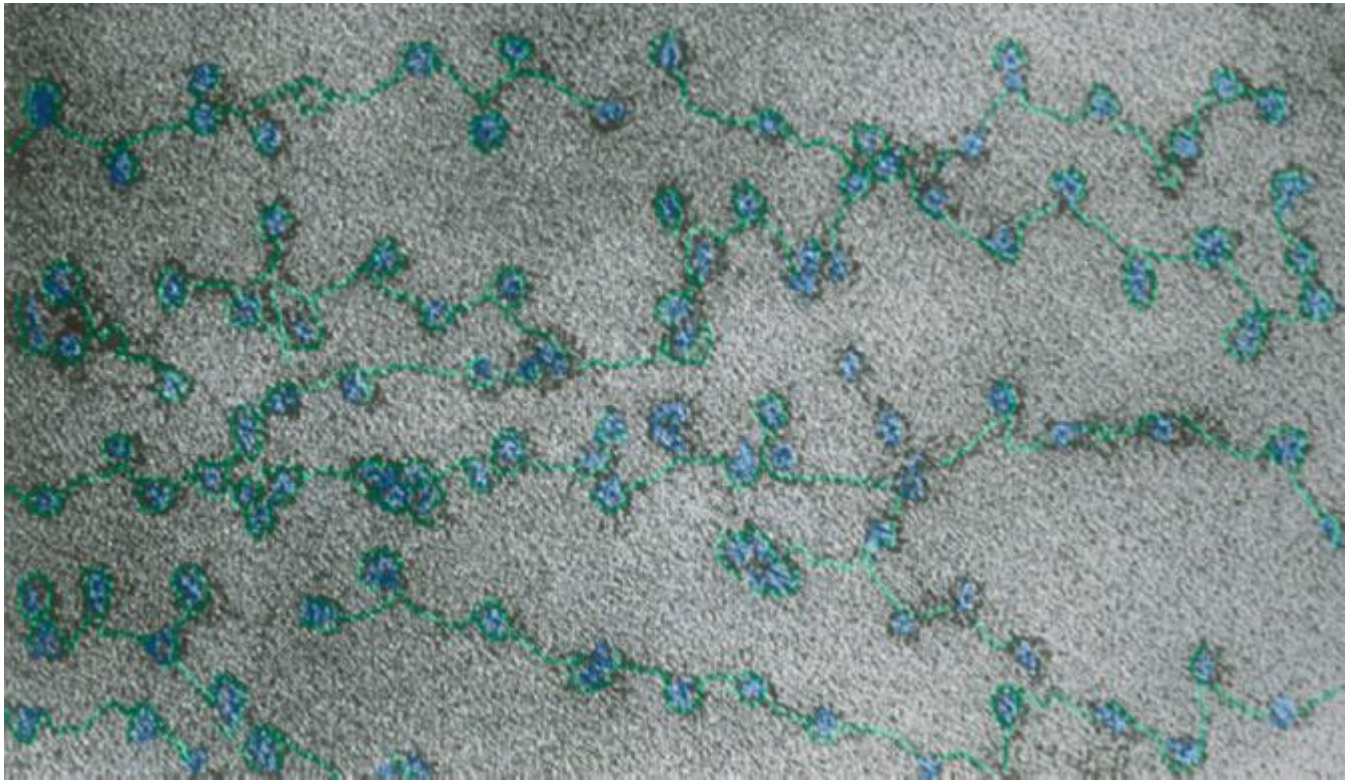
- Non-coding regions <u>recruit</u>:
  - Transcription factors
  - Activators
  - miRNAs, lncRNAs (types of ncRNAs)
  - Accessory proteins

(non-exhaustive list)

1) Chromatin-level regulation: Nucleosome packaging

“Beads on a string”

Electron microscope image of nucleosomes and DNA (with false-color added)



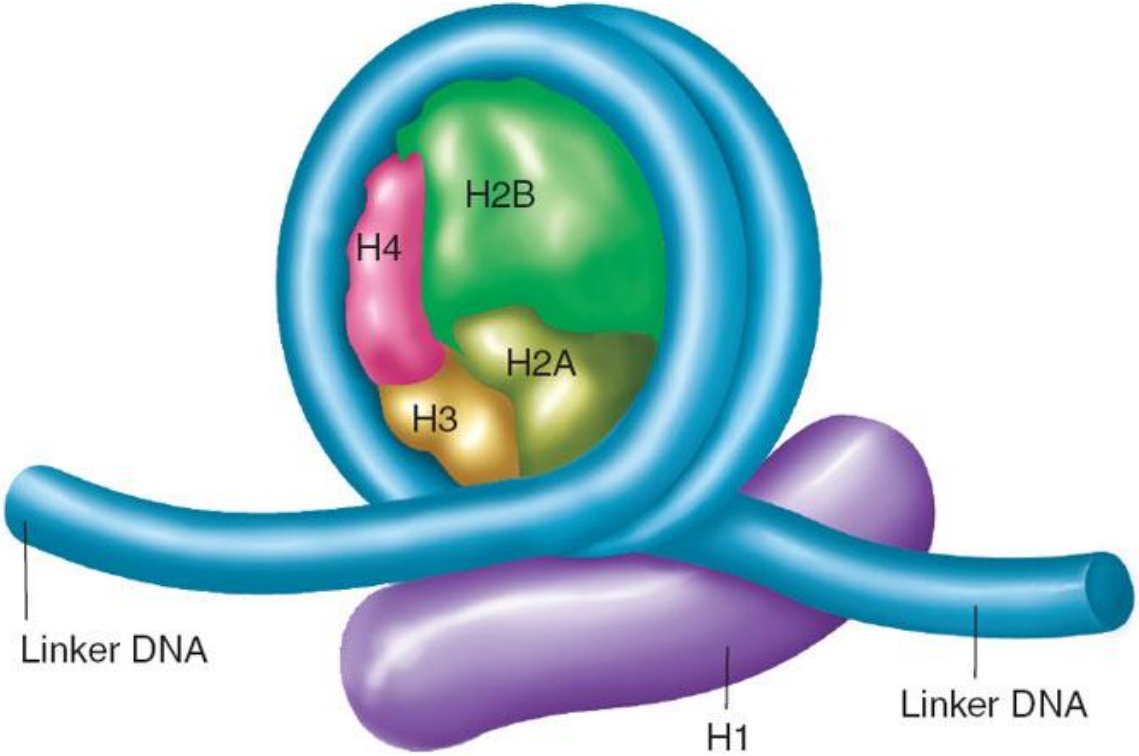
Nucleosome structure

Histone proteins: typically H2A, H2B, H3, H4, and H1

Linker DNA: pieces of DNA between nucleosomes

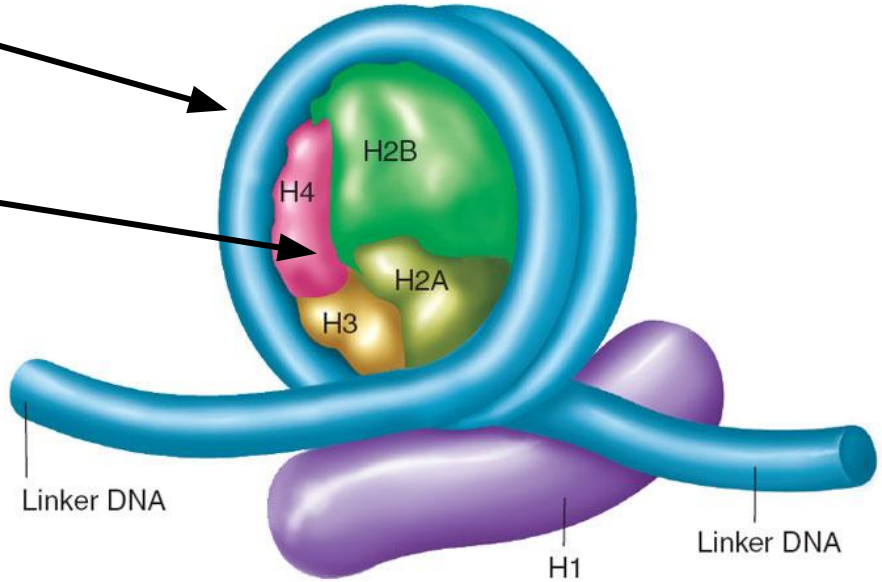
160 bp of DNA sequence wraps twice around the histone octamer

An additional 40bp are from the linker DNA



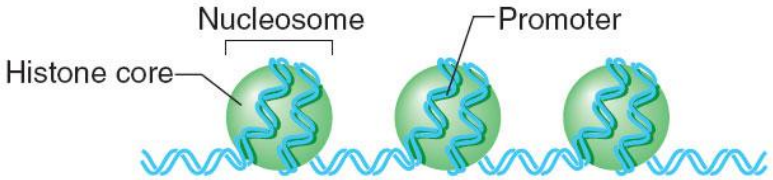
Nucleosomes can regulate genes in two ways

1. Nucleosome packaging
2. Histone tail modifications

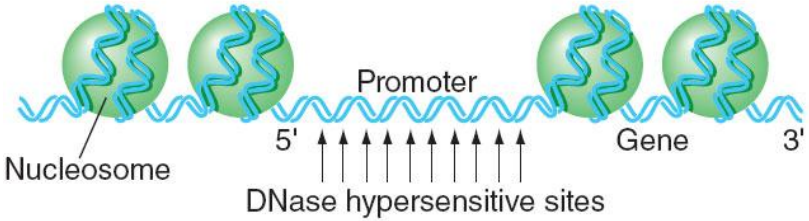


Nucleosome packaging and gene expression

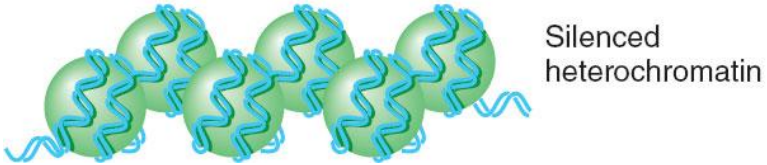
(a) Promoters are hidden when wrapped in nucleosomes.



(b) Chromatin remodeling complexes can expose gene promoters.



(c) Nucleosomes in heterochromatin are tightly packed.



DNase hypersensitive sites: If DNA is exposed (accessible), they are sensitive to cutting via a DN<u>ase*</u> enzyme

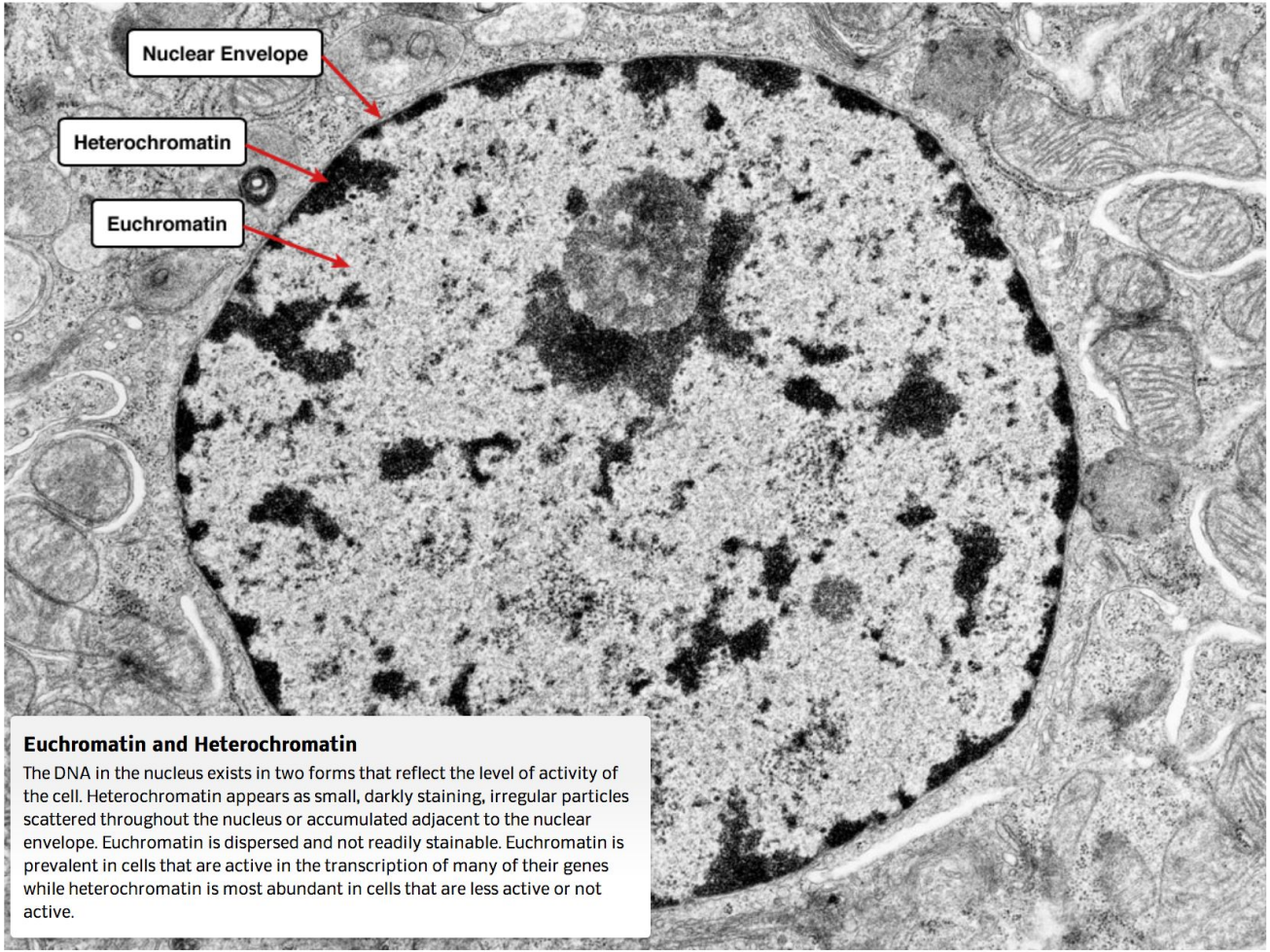
*General rule: if something ends in “ase” (pronounced: ace), then it is usually an enzyme

The Google Earth view of genomic gene expression:

Euchromatin - condenses/decondenses throughout the cell cycle (eu~true chromatin)

Heterochromatin - remains condensed throughout (coined in 1928 by Emil Heitz)

Note: Heterochromatin is darker than euchromatin because it is heavily stained (by uranyl acetate) due to its high protein (histone) content

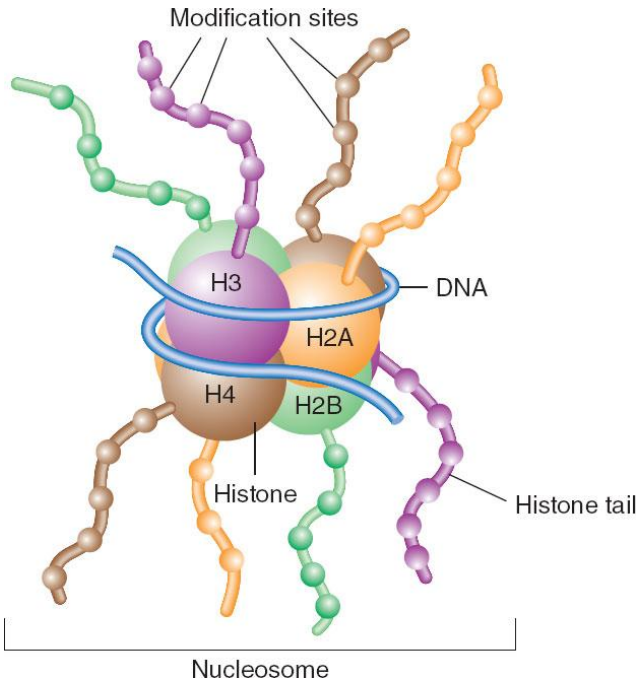


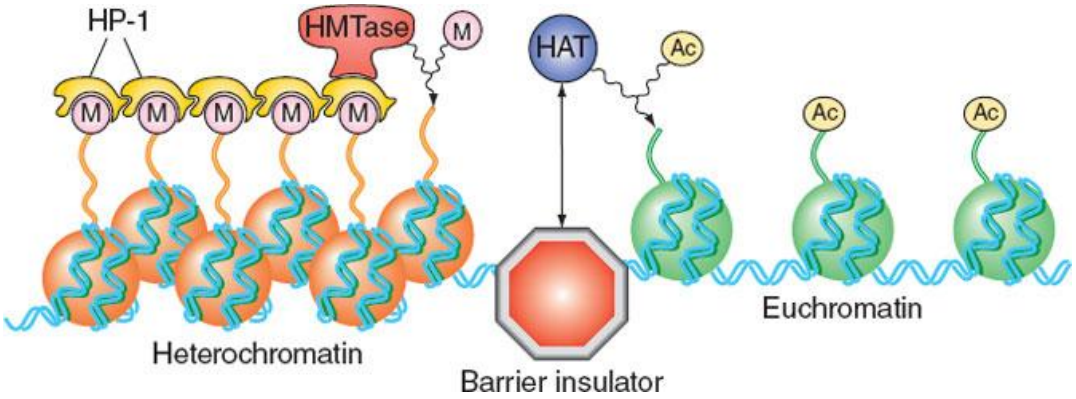
Euchromatin and Heterochromatin

The DNA in the nucleus exists in two forms that reflect the level of activity of the cell. Heterochromatin appears as small, darkly staining, irregular particles scattered throughout the nucleus or accumulated adjacent to the nuclear envelope. Euchromatin is dispersed and not readily stainable. Euchromatin is prevalent in cells that are active in the transcription of many of their genes while heterochromatin is most abundant in cells that are less active or not active.

http://medcell.med.yale.edu/histology/cell_lab/euchromatin_and_heterochromatin.php

Modification of histone tails marks heterochromatin and euchromatin regions



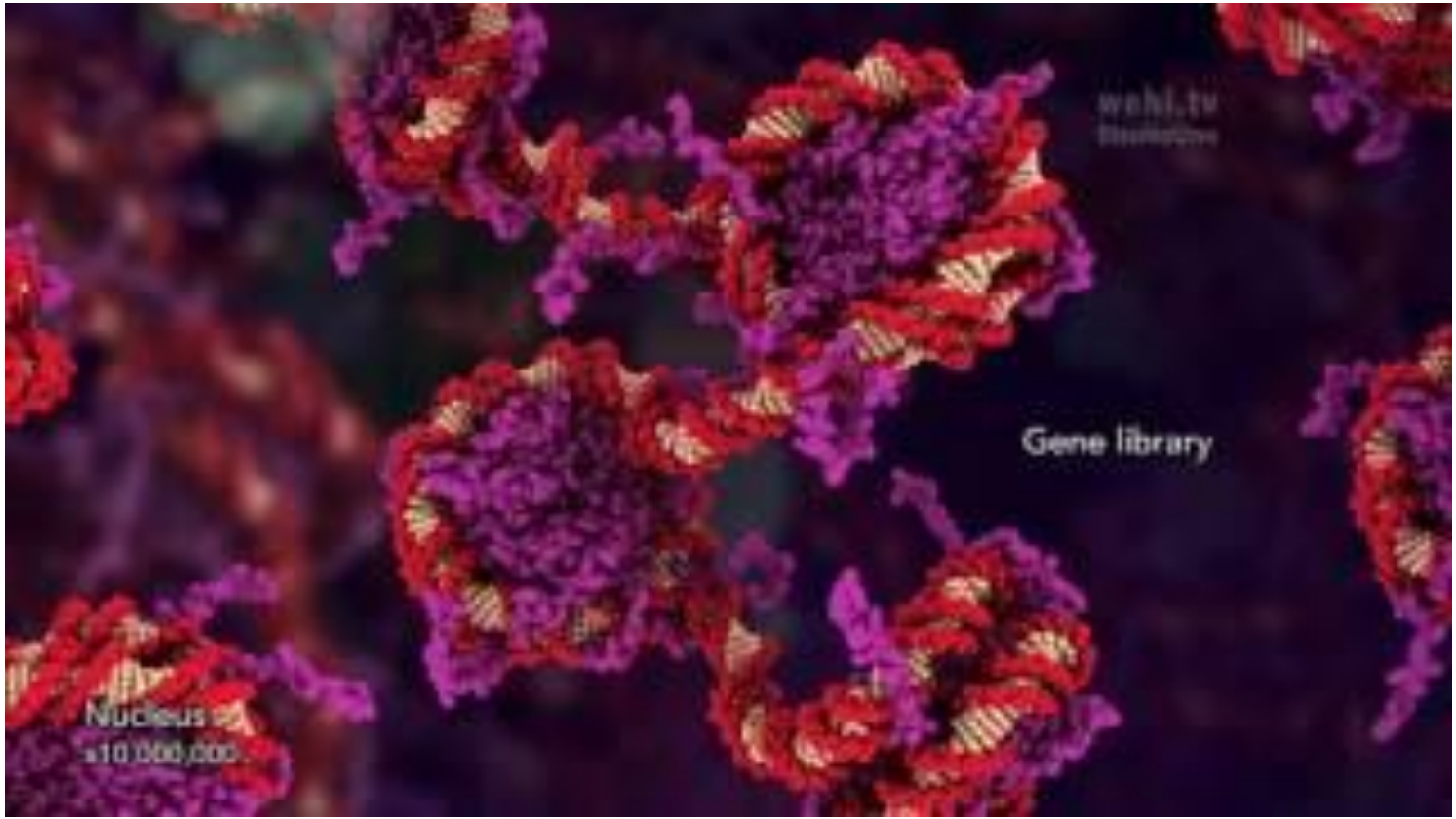


M - methylation (~closed); Ac - acetylation (~open)

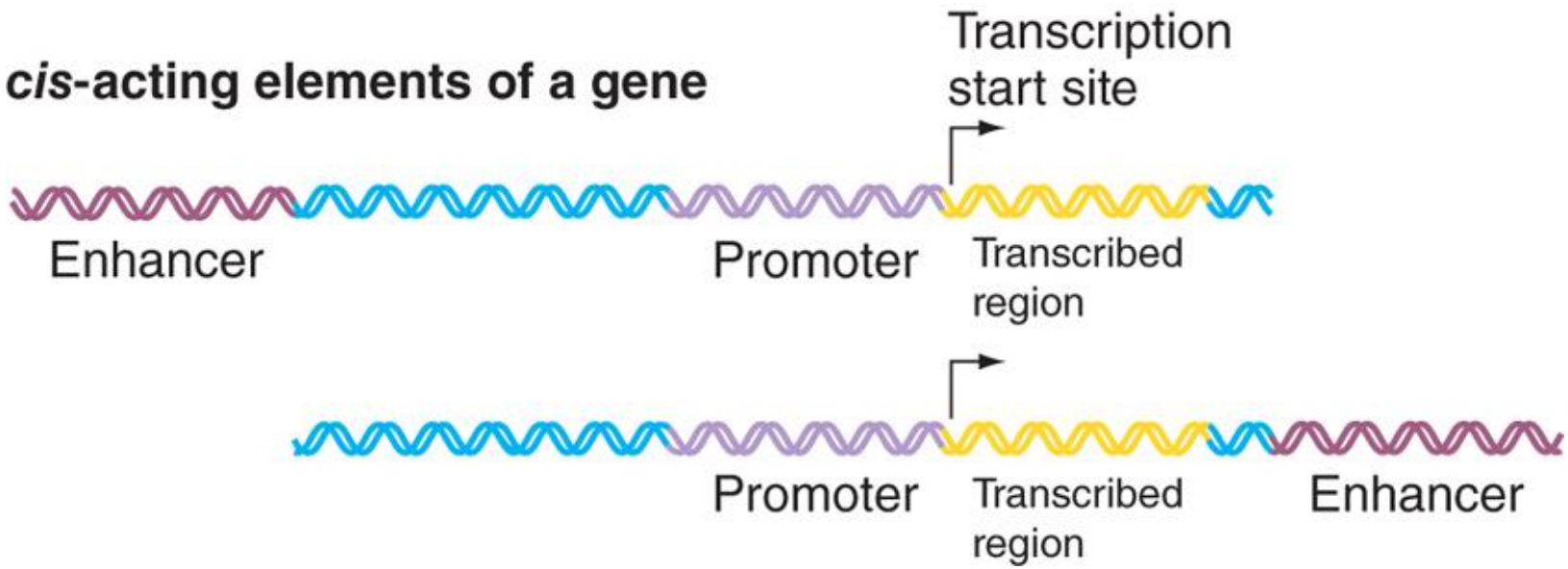
HMTase - Histone methyltransferase
HAT - Histone acetyltransferase

HP1 - binds methylated tails and recruits more HMTase

Chromatin organization

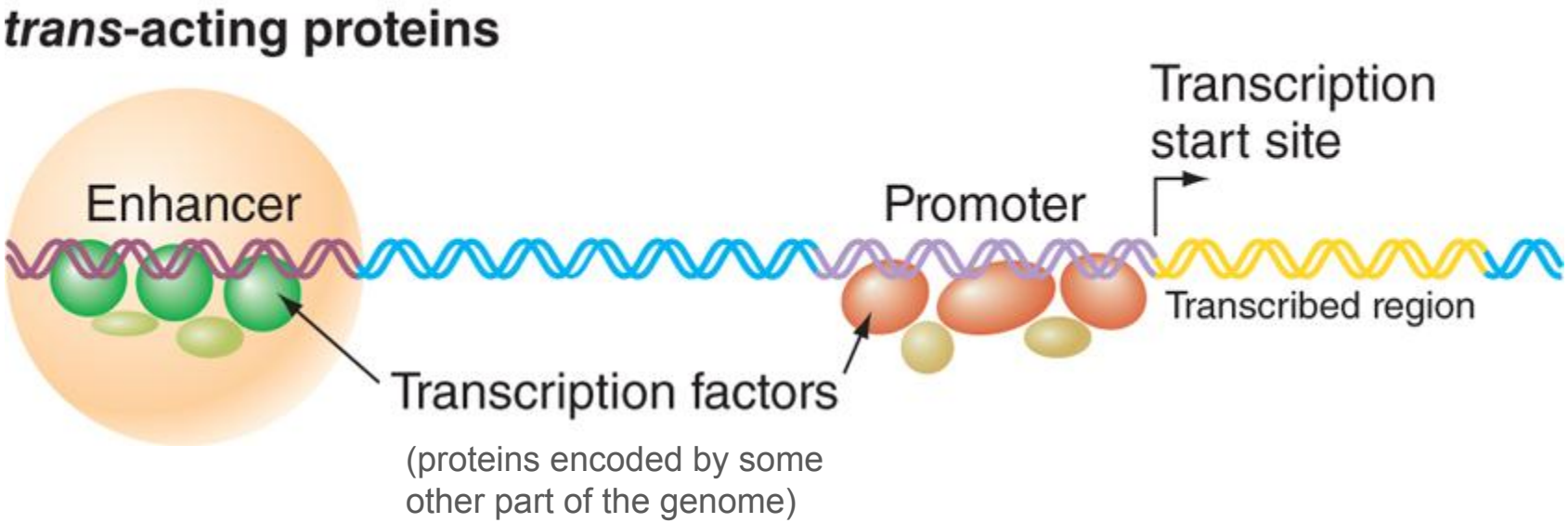


2) Transcriptional-level control of gene regulation through cis- and trans-factors

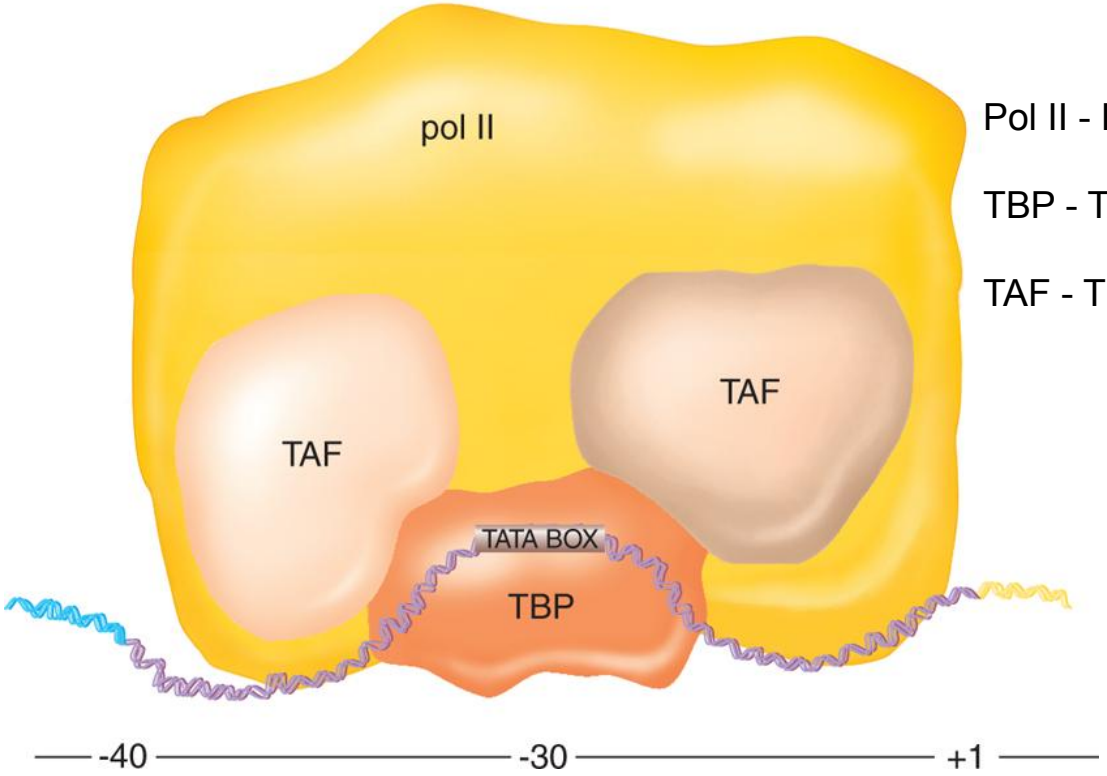


Enhancers can be upstream or downstream of the promoter

Control of gene regulation through cis- and trans-factors



Basal factors bind to promoters of genes and are the minimum set of proteins needed for transcription



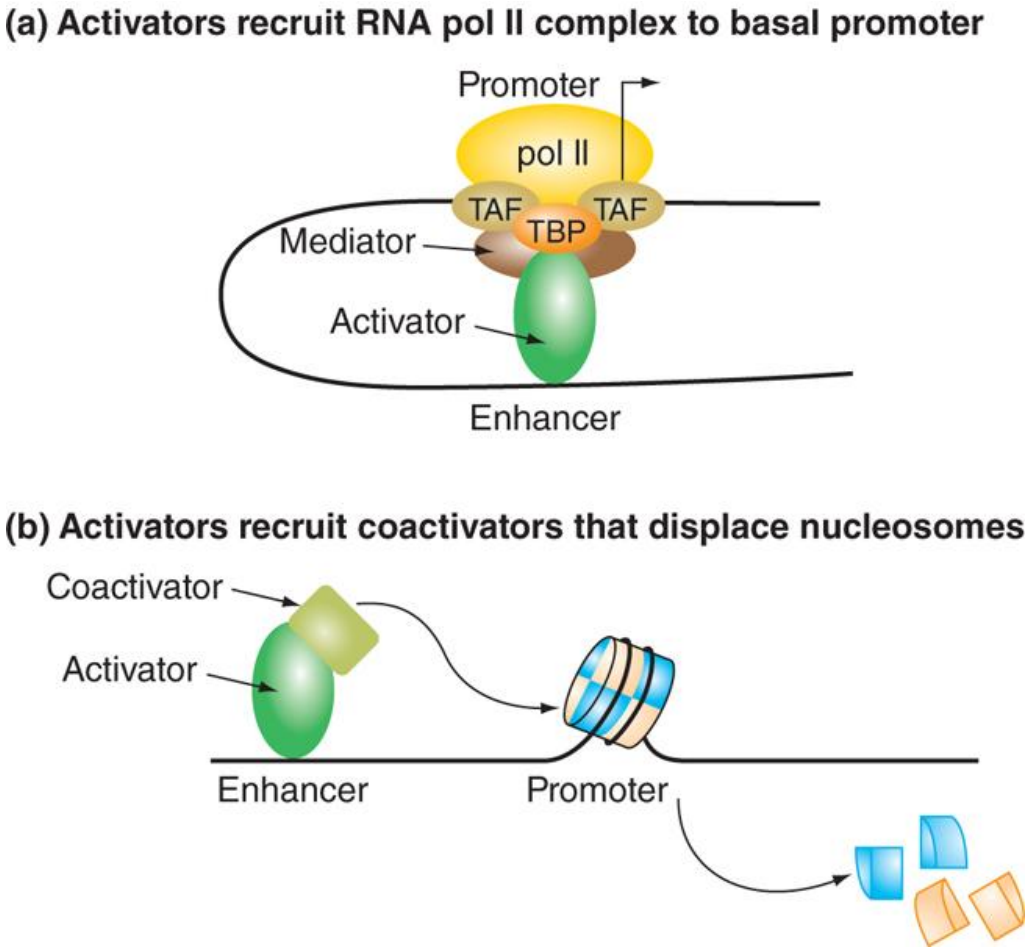
Pol II - RNA polymerase II

TBP - TATA box-binding protein

TAF - TBP-associated factors

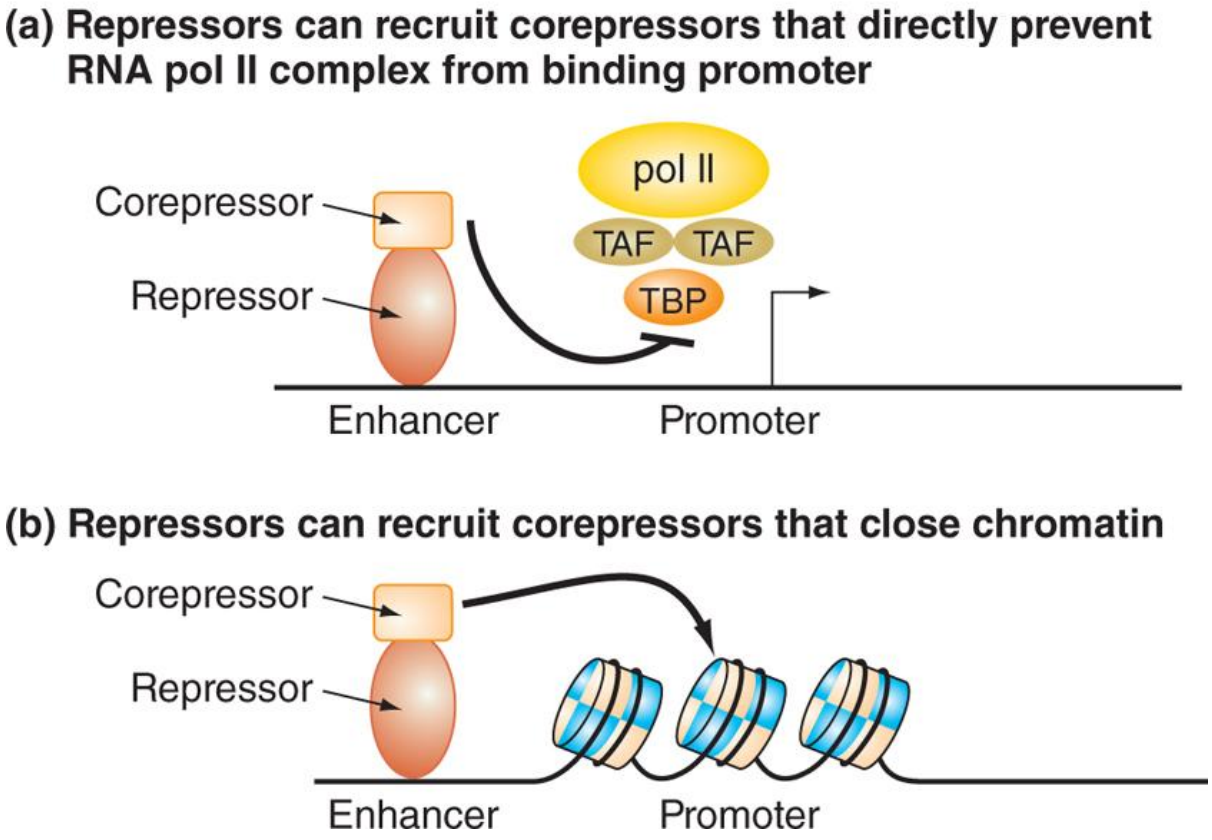
What transcriptional activators do

- Start transcription
- Facilitate the opening of chromatin



What transcriptional repressors do

- Block transcription
- Facilitate the closing of chromatin

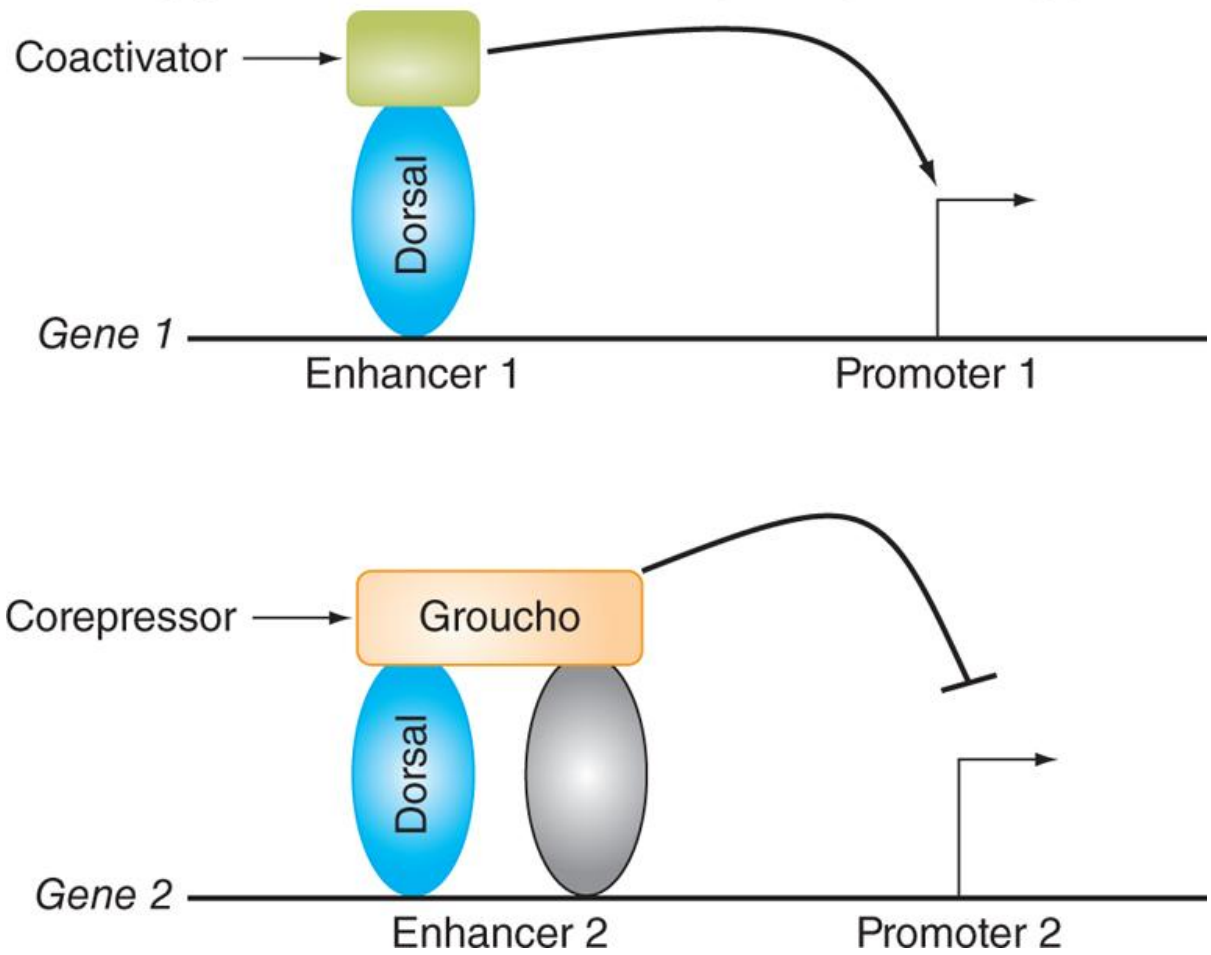


A transcription factor can be both an activator and a repressor

Depends on other co-factors/co-regulators:

Other cofactors need to be expressed (turned on)

Depends on the presence of binding site for co-factors

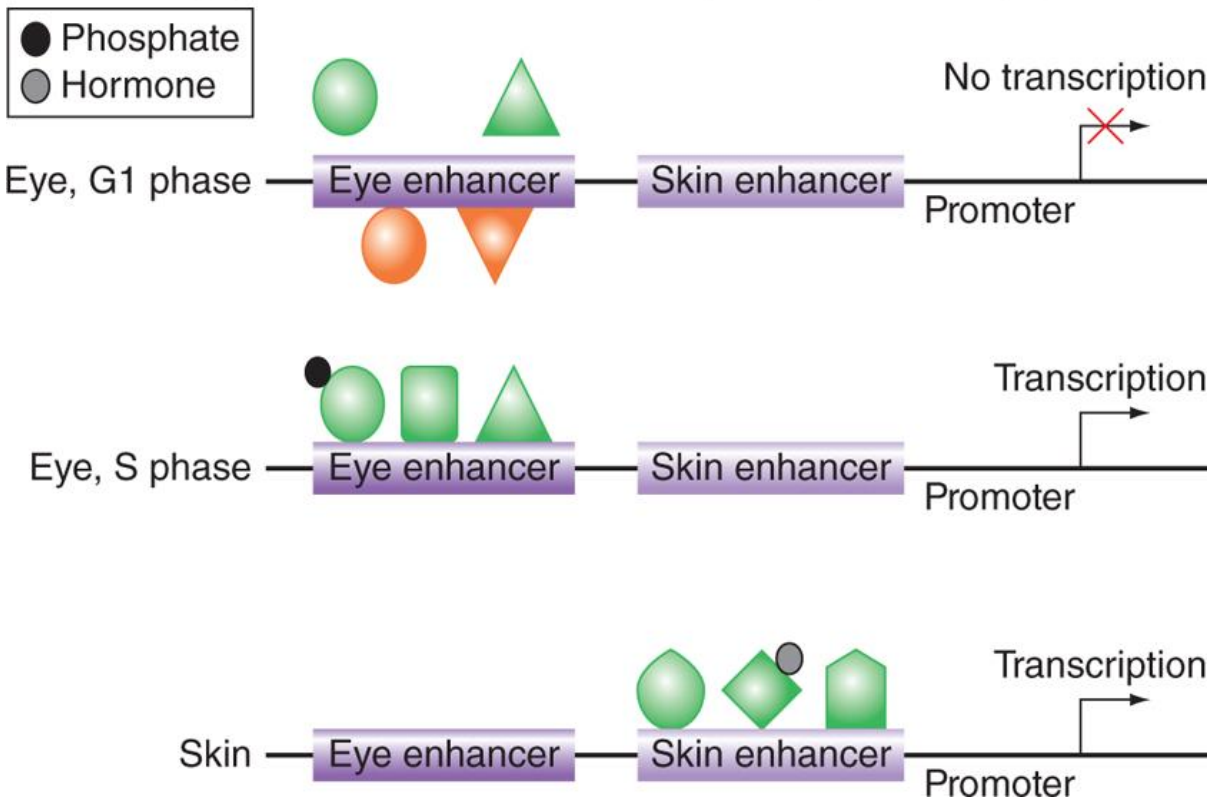


Enhancers govern tissue- and temporal-specific gene expression

Green proteins - activators

Orange proteins - repressors

Genes can have multiple enhancers that regulate its expression under different conditions

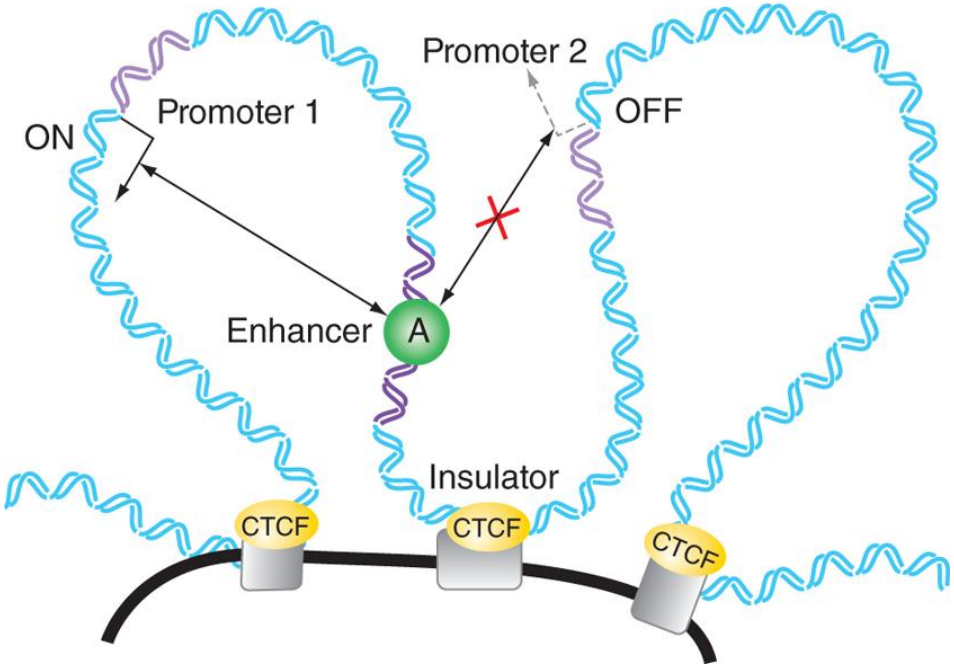


Insulators control enhancer-promoter interactions

RFP and *GFP* transcribed

Only *GFP* transcribed

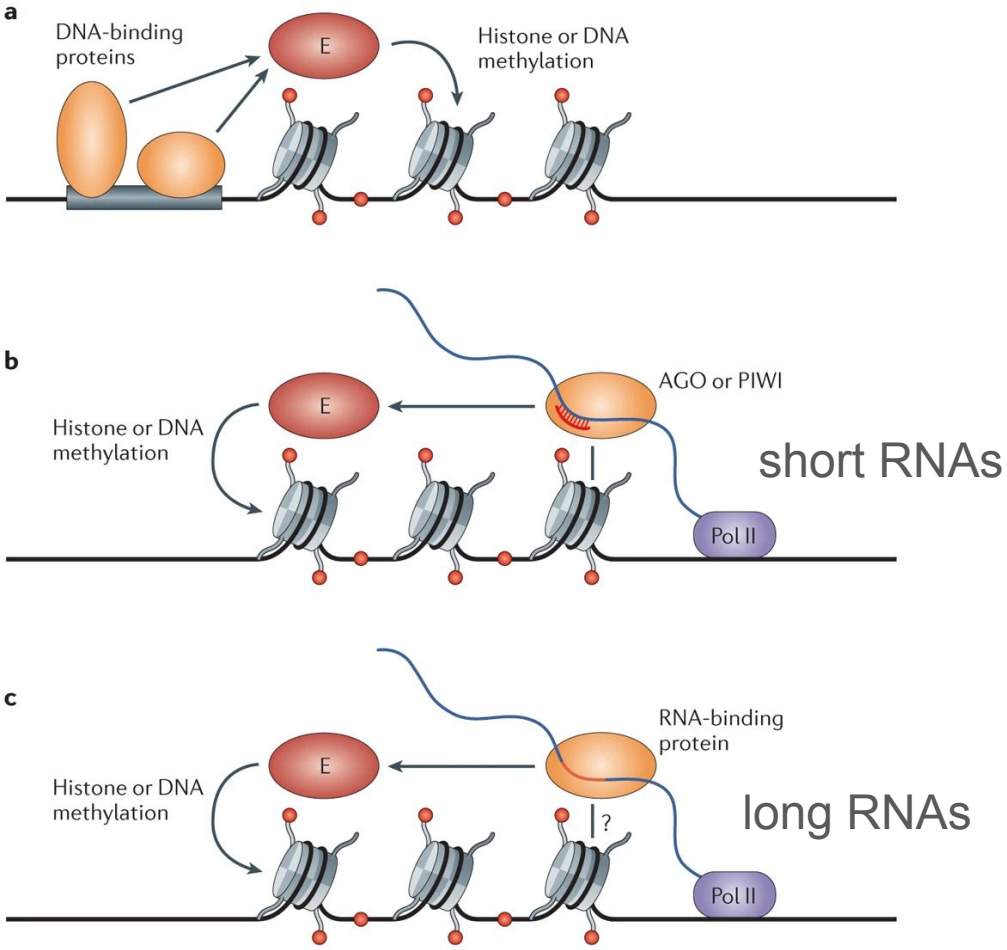
(Enhancer now in separate loop from promoter 2)



CTCF = zinc finger transcription factor that binds to insulator regions that associate with the nuclear membrane

How do ncRNAs fit in?

ncRNAs are an alternative to DNA-binding proteins as <u>specificity determinants for epigenetic regulation</u> of gene expression



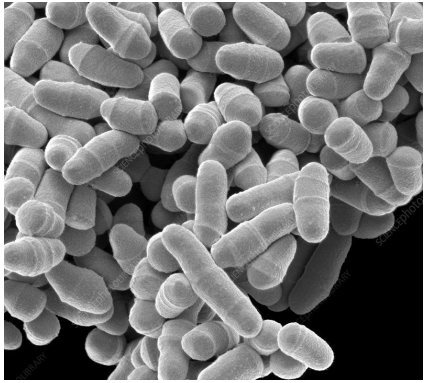
Holoch and Moazed 2015

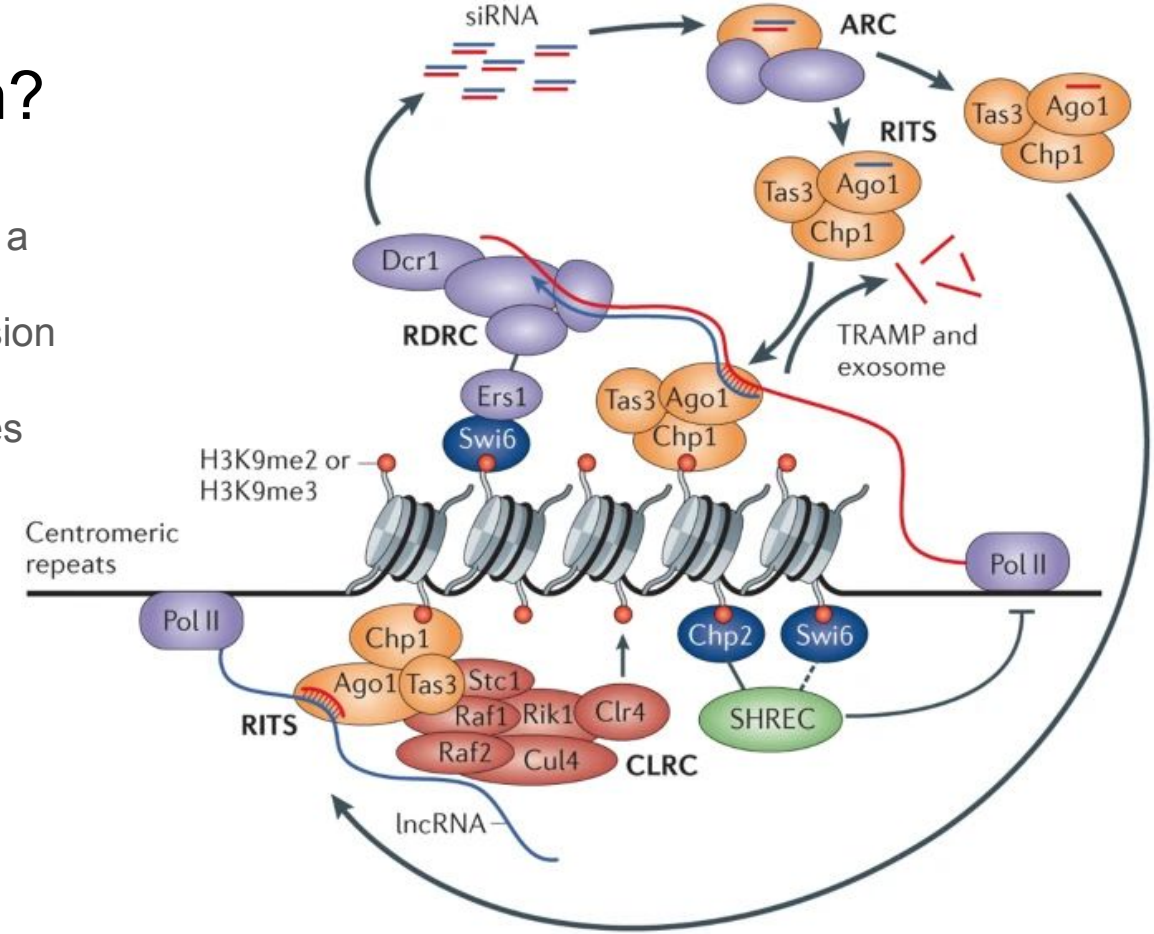
Nature Reviews | Genetics

How do ncRNAs fit in?

The 'nascent transcript' model and a self-reinforcing epigenetic loop in *Schizosaccharomyces pombe* (fission yeast)

→ Heterochromatin expresses siRNAs for its own silencing



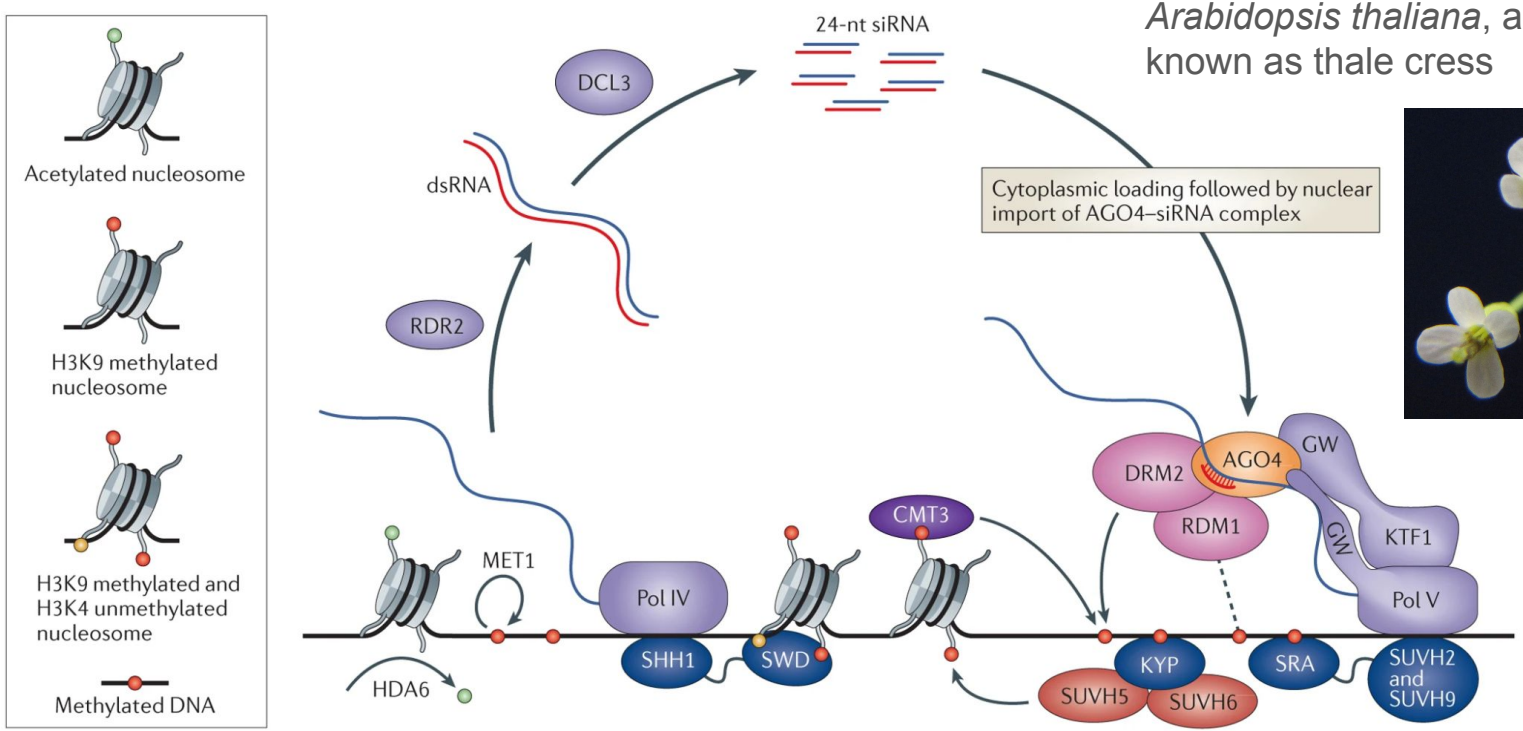


Nature Reviews | Genetics

How do ncRNAs fit in?

A self-reinforcing loop linking siRNAs to DNA and histone methylation in *A. thaliana*

Arabidopsis thaliana, also known as thale cress



Nature Reviews | Genetics

How do we identify these regulatory regions?

Genomics project to identify epigenomic features

How do we identify these regulatory regions?

Genomics project to identify epigenomic features

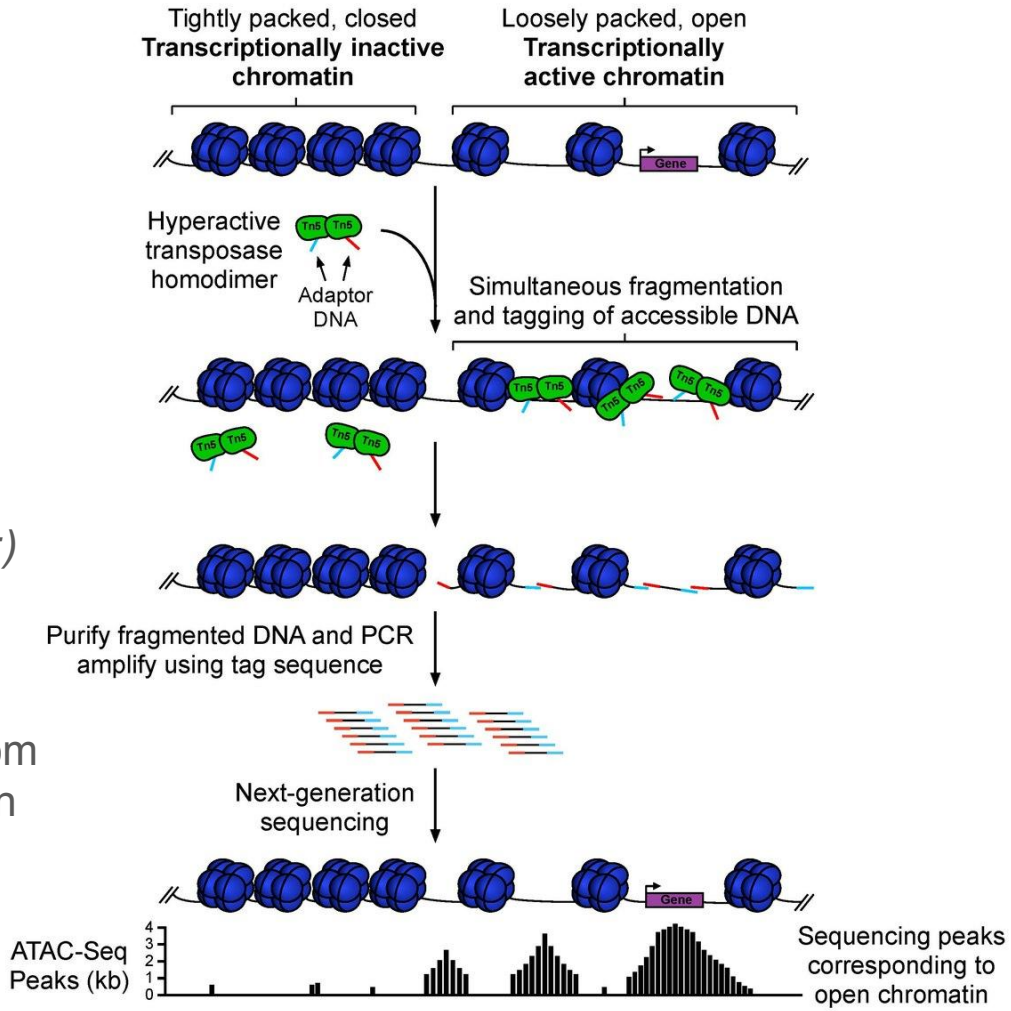
1) Open vs closed DNA → ATACseq
2) Protein-bound DNA → ChIPseq
3) DNA in physical contact → Hi-C

1) ATAC-Seq

⇒ Assay for Transposase-Accessible Chromatin with sequencing

ATACing open chromatin… :)

*The reads you get back [from sequencing] map to the open regions of the genome:



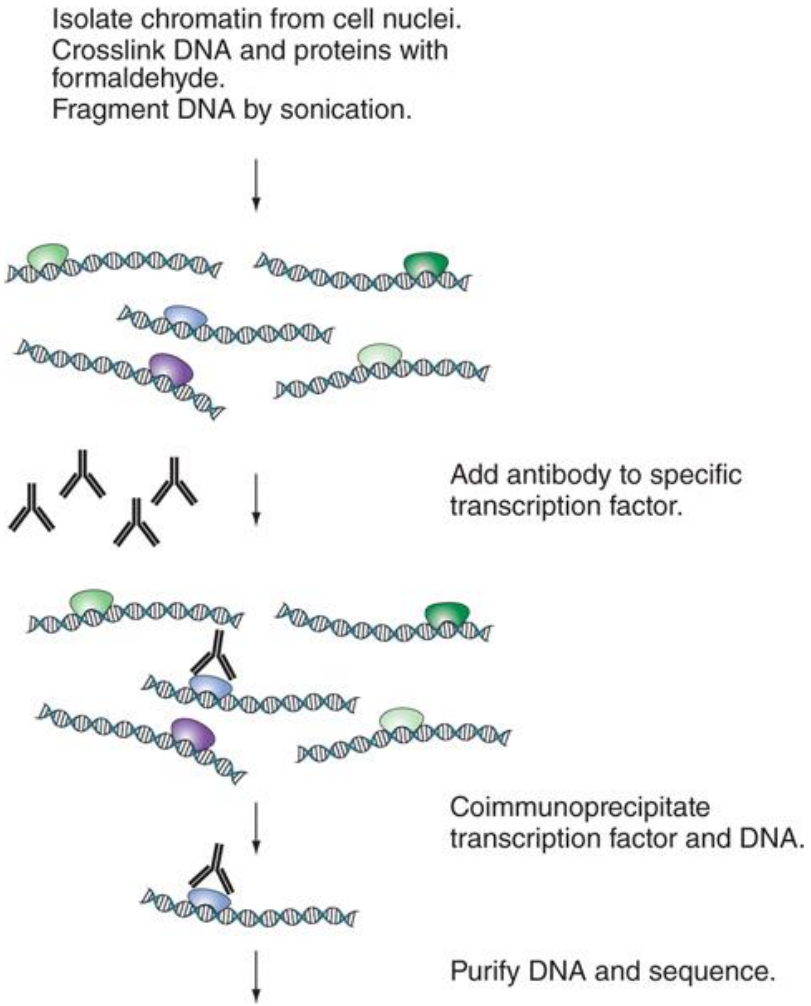
Wikipedia

2) ChIP-Seq

Ch ⇒ chromatin

IP ⇒ immunoprecipitation

*The reads you get back [from sequencing] map to regions of the genome that bound to the TF protein



3) Chromatin contact sequencing: 3C, Hi-C, Micro-C, etc.

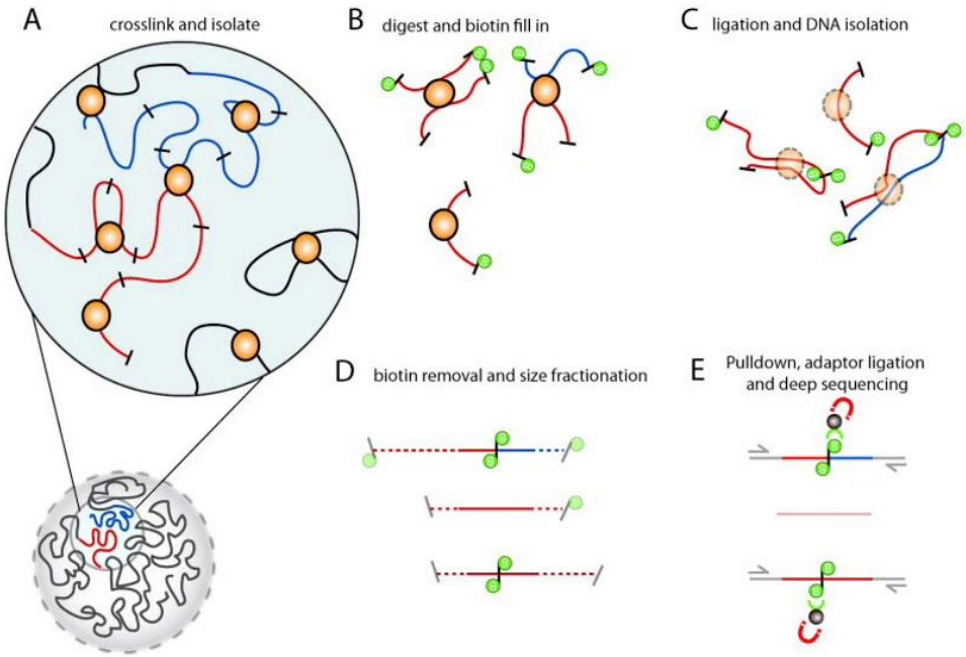
3C == Chromatin Conformation Capture

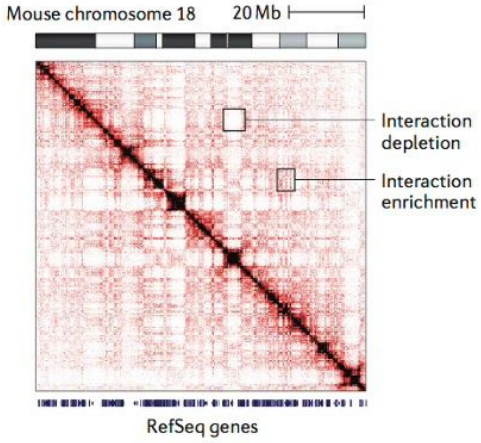
X-C are derivations of the original protocol

*The reads you get back [from sequencing] are from DNA that was physically touching → chimeras reveal trans- and long-range cis-interactions

Hi-C: all-by-all interactions

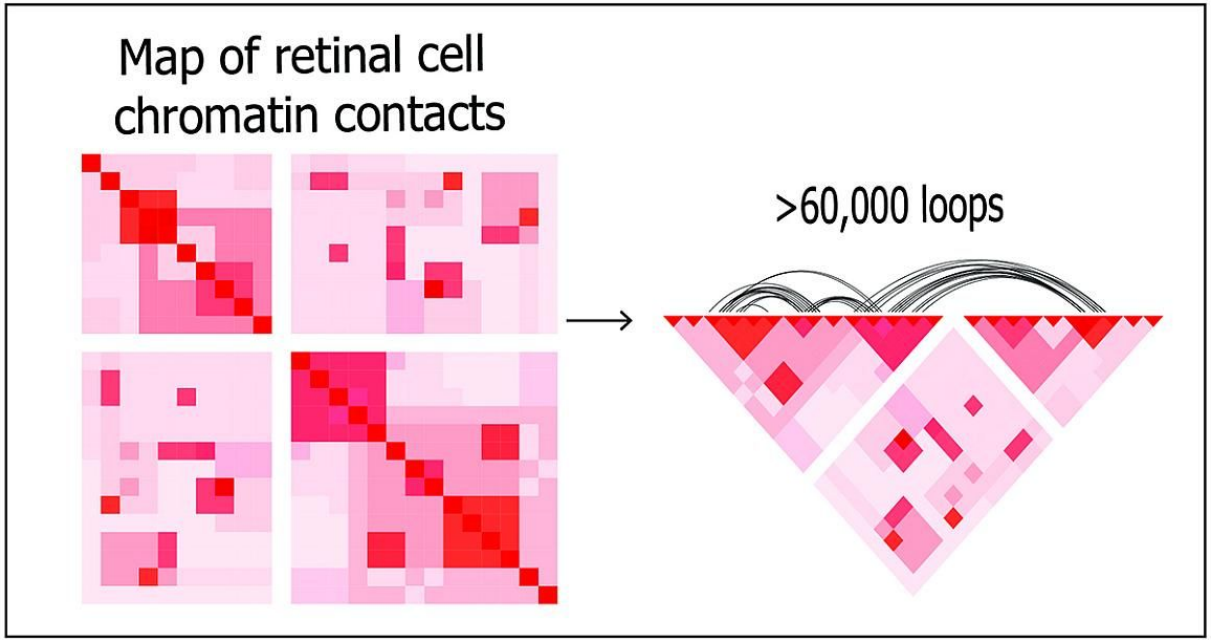
- **Note: Biotin labels fragment ends without interfering with folding of bound proteins, DNA**





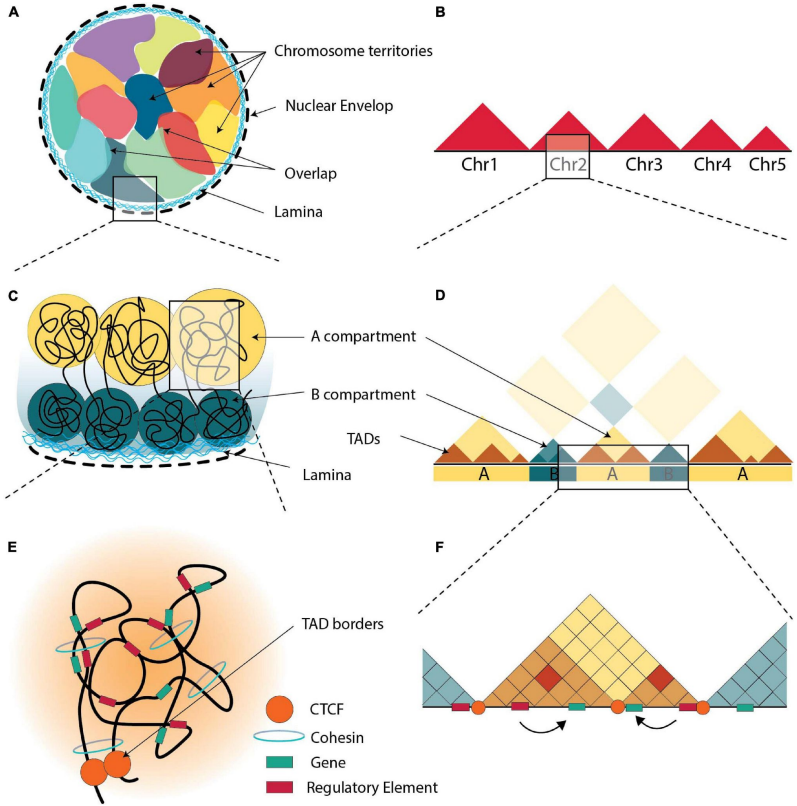
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3874846/

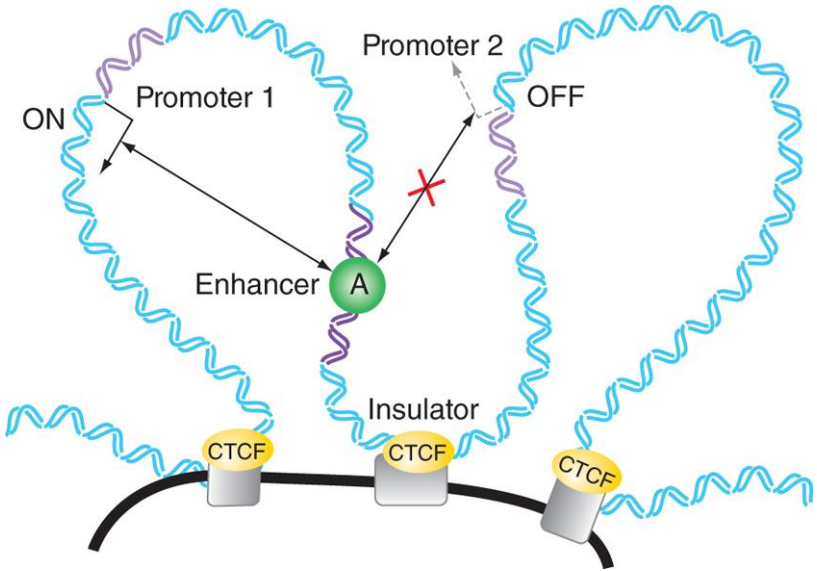
3) Chromatin contact sequencing: 3C, Hi-C, Micro-C, etc.



Marchal et al. 2022; Anand Swaroop, Ph.D., NEI

3) Chromatin contact sequencing: 3C, Hi-C, Micro-C, etc.

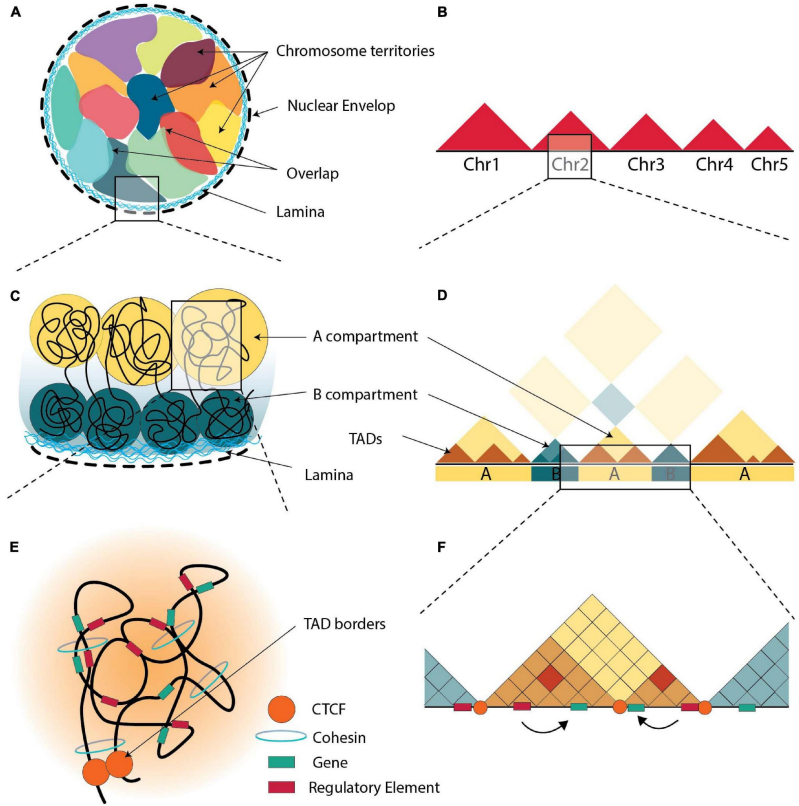




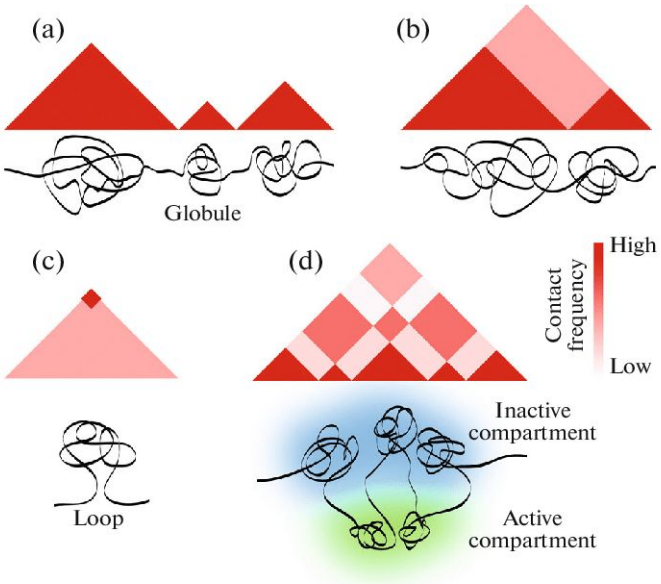
TADs = Topologically associating domains
CTCF = Zinc finger transcription factor

Boltsis et al. et al. 2021

3) Chromatin contact sequencing: 3C, Hi-C, Micro-C, etc.



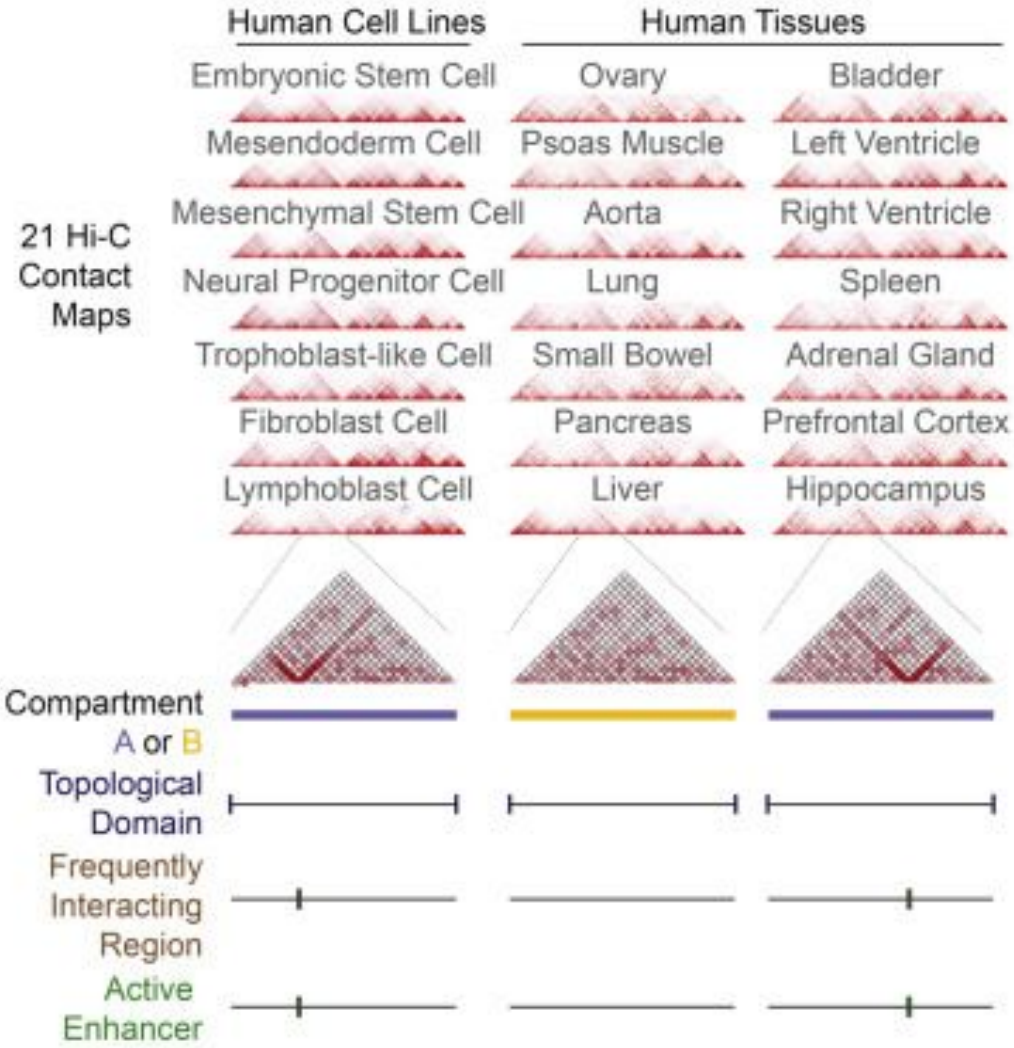
Boltsis et al. et al. 2021



Razin et al. et al. 2019

TADs = Topologically associating domains
CTCF = Zinc finger transcription factor

Genome-wide chromatin contact sequencing reveals tissue-specific epigenetic chromatin compartmentalization



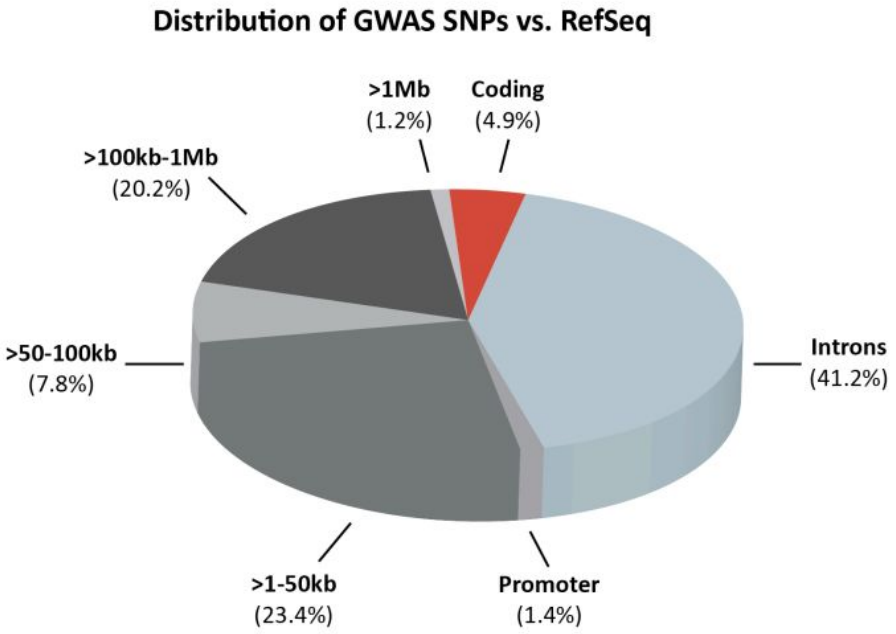
Schmitt et al. et al. 2016

Most human traits and diseases are associated with SNPs in non-coding regions

Identifying cis-acting regulatory elements in the genome can aid in interpreting the functional effect of variants in non-coding regions

However, simply knowing that a SNP overlaps a regulatory region <u>does not indicate that it is a causal</u> SNP for a disease/phenotype.

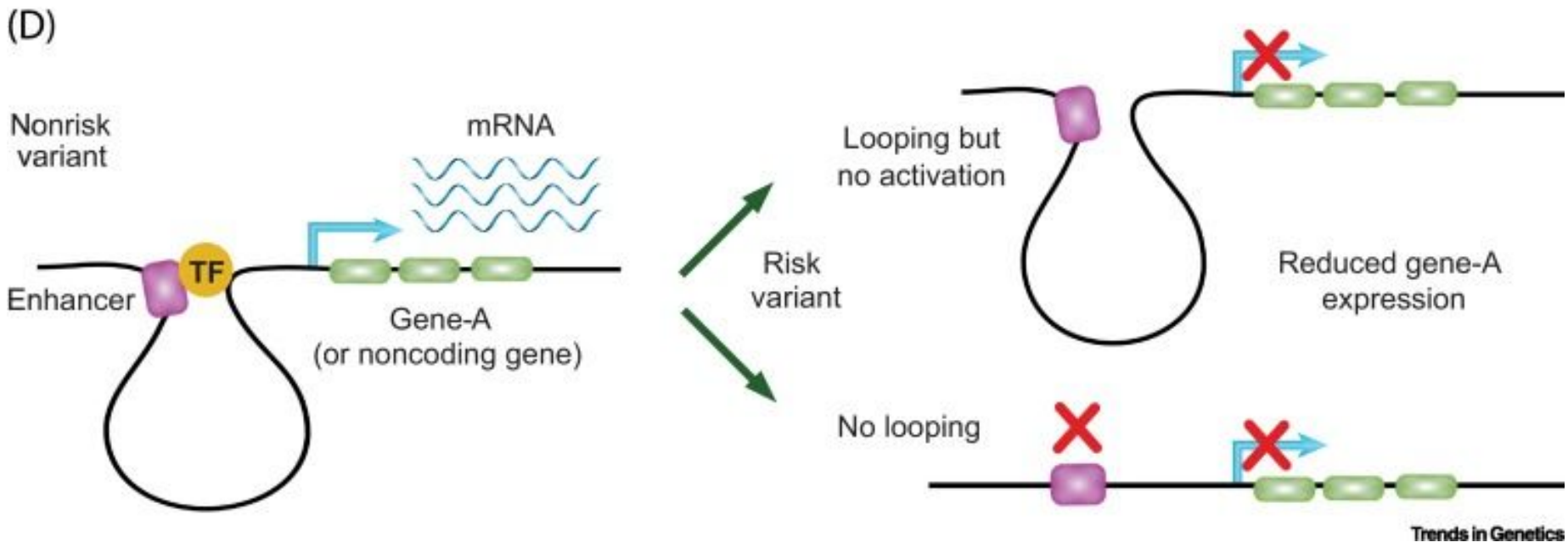
Further studies (e.g. cell line experiments, animal models) need to be performed to determine if the SNP is possibly a causal SNP.



Maurano et al. Science 2012

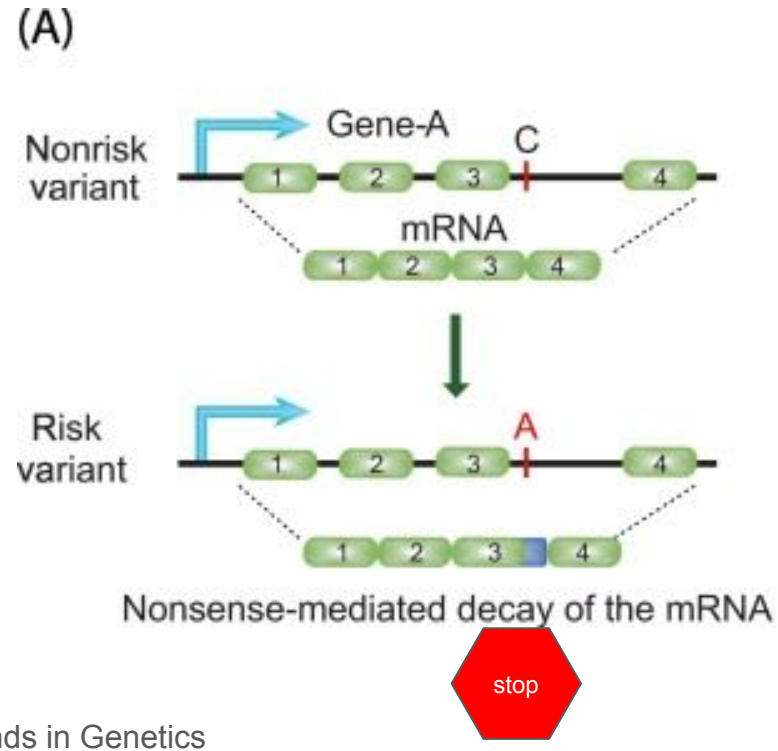
Some examples of non-coding variation on gene expression…

Risk variants can alter TF binding



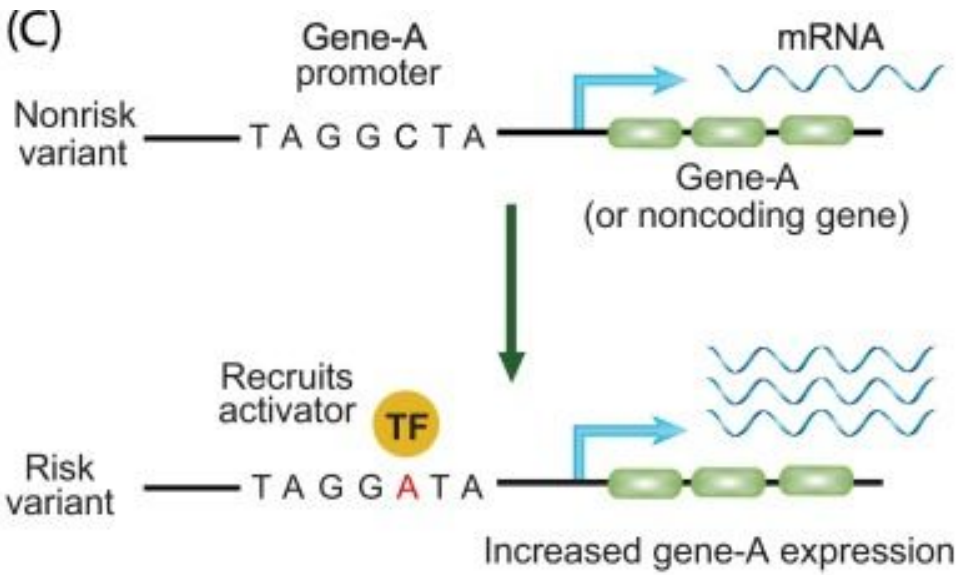
French and Edwards. 2020. Trends in Genetics

Risk variants can activate cryptic splice acceptor sites



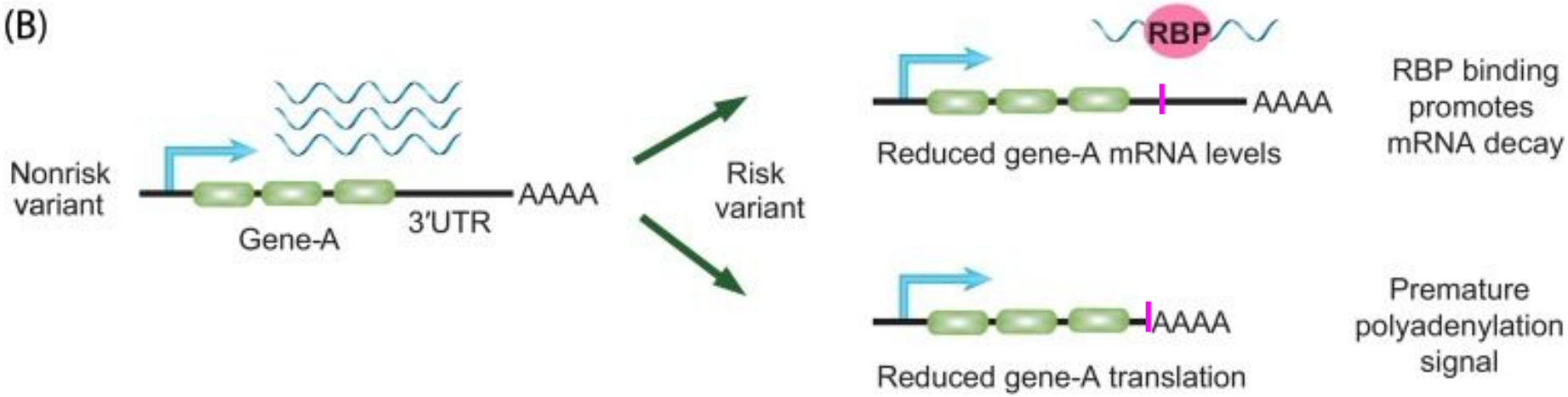
French and Edwards. 2020. Trends in Genetics

Risk variants in promoter regions can lead to the recruitment of additional transcription factors (TFs)



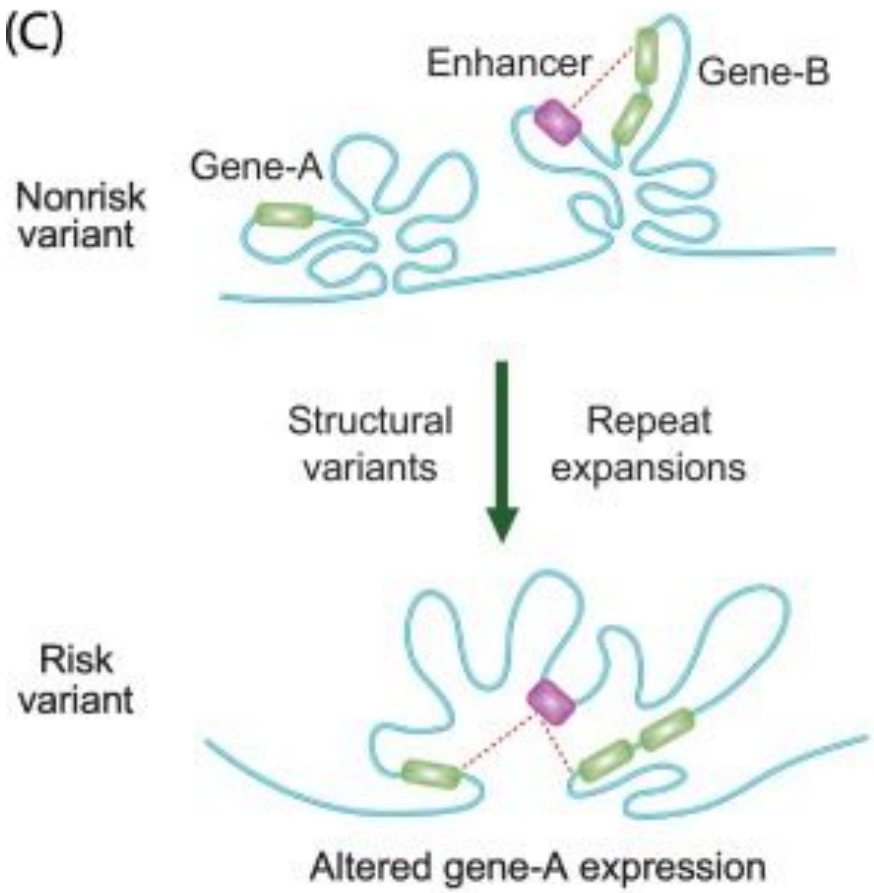
French and Edwards. 2020. Trends in Genetics

Risk variants in 3′UTRs can create an RNA-binding protein (RBP) recognition site



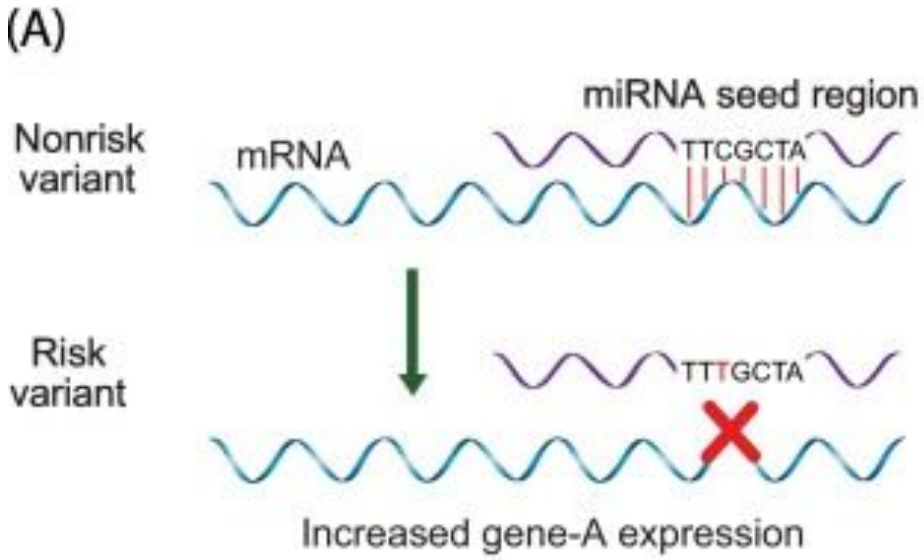
French and Edwards. 2020. Trends in Genetics

Structural variants or repeat expansions can disrupt TAD boundary elements



French and Edwards. 2020. Trends in Genetics

Risk variants in miRNA seed regions can prevent miRNA binding to target genes

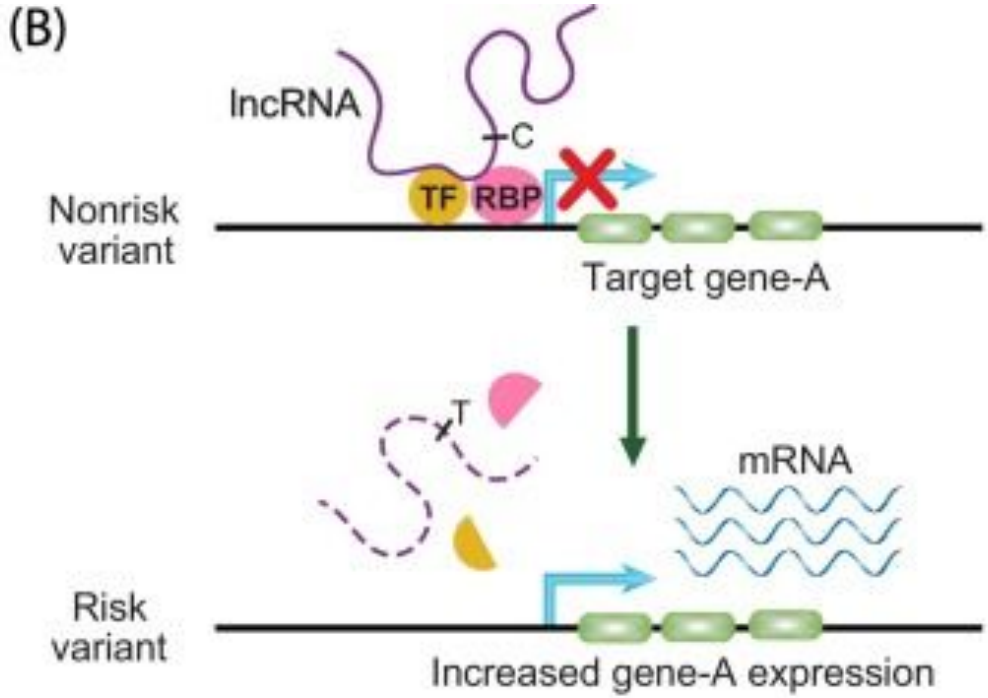


French and Edwards. 2020. Trends in Genetics

Risk variants can prevent protein binding to lncRNA

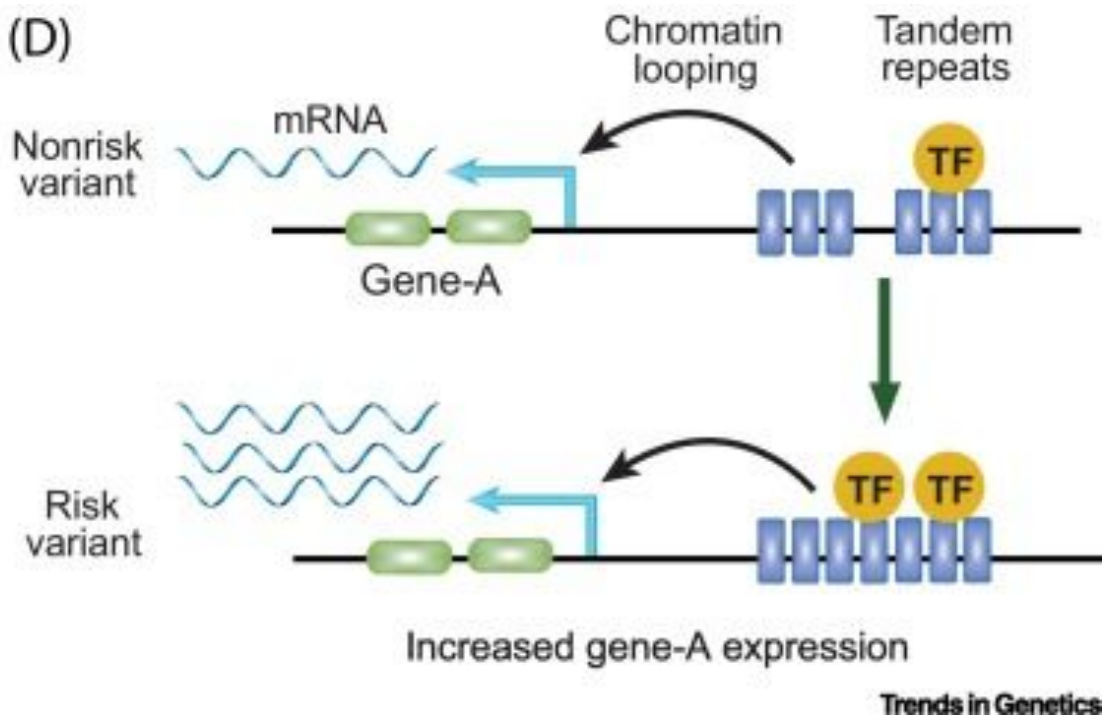
TF= transcription factor

RBP=RNA-binding protein



French and Edwards. 2020. Trends in Genetics

Risk variants can connect adjacent tandem repeats



French and Edwards. 2020. Trends in Genetics

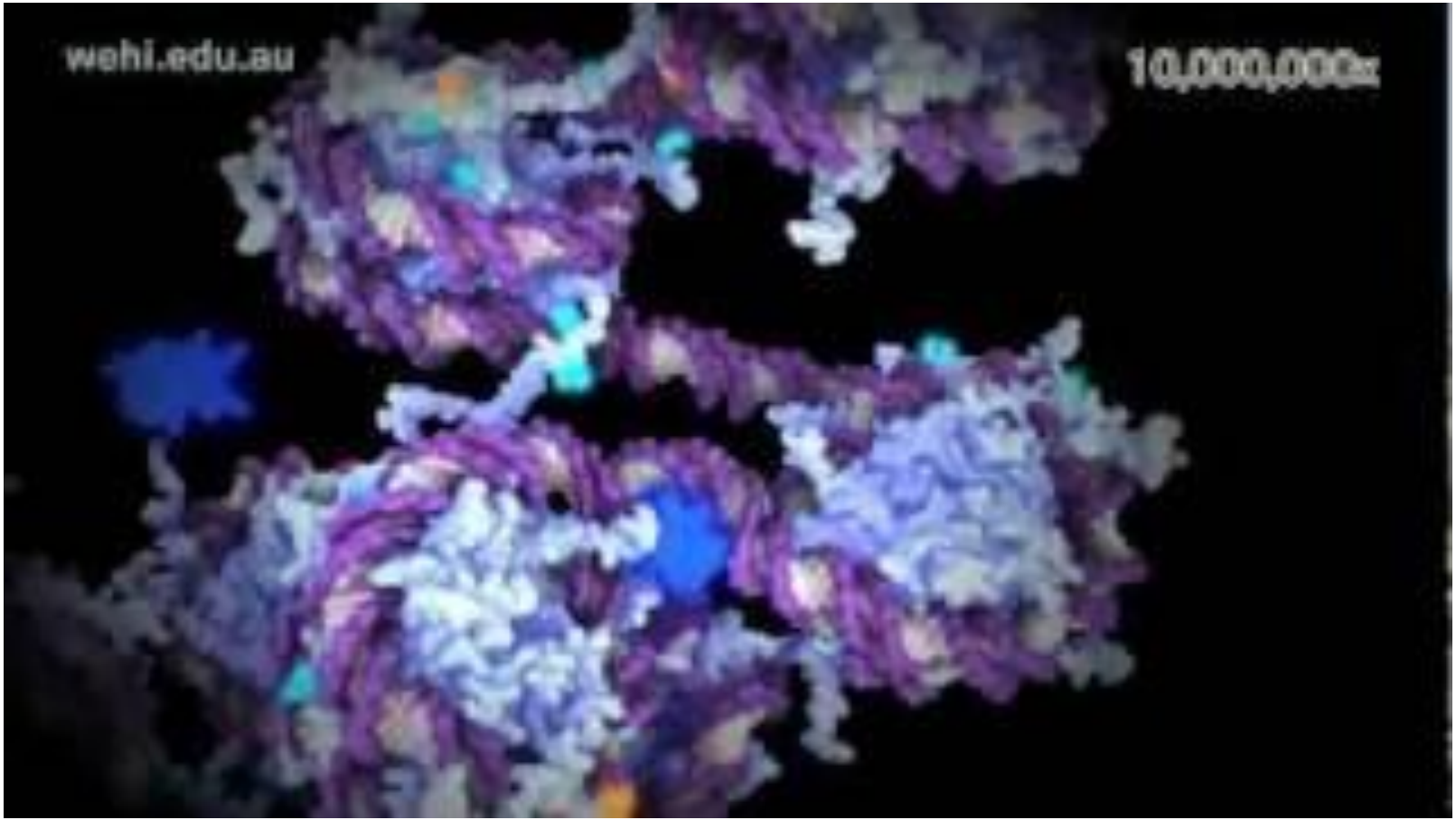
For next class:

Start working on Problem Set 14: Gene regulation and epigenetics

Continue reading in Genetics: From Genes to Genomes

a. Finish Chapter 12.1-12.3, Chapter 17.1, 17.2
b. Start Chapter 17.3

What have chromatin contact maps taught us?

Rao et al. 2014. Cell

Epigenetics: X-inactivation